CAMBRIDGE IGCSE · 0620 · EXTENDED

Chemistry

Complete notes — all 12 topics + practical skills

How to use this

1. Read one numbered section (they match the SaveMyExams sub-topics).
2. Do the SaveMyExams topic questions for that section and mark them with the mark scheme.
3. Turn every mistake into a flashcard. Aim for about 75–80%, then move on.
4. Before moving to the next topic, tick the self-check list at the end of the topic.

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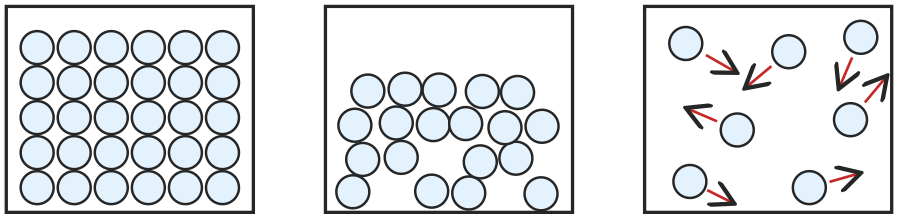
Topic 1 States of Matter

Cambridge IGCSE Chemistry 0620 · Extended · Section 1 matches the SaveMyExams sub-topic, so you can do their questions after it

1 Solids, Liquids & Gases

Kinetic theory

> ***The kinetic particle theory***
> All matter is made of **tiny particles** (atoms, molecules or ions) that are always moving. The higher the temperature, the more **kinetic energy** the particles have and the faster they move. The state of a substance depends on how strong the **forces of attraction** between its particles are compared with how much energy the particles have.

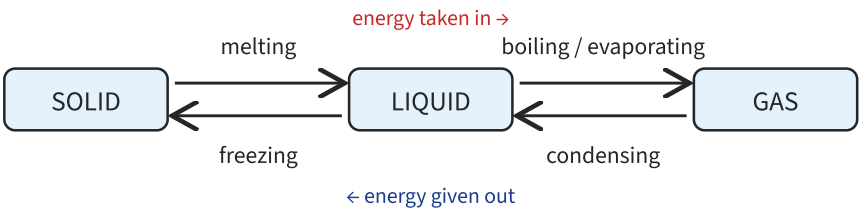


Particle diagrams — exam diagrams need the right spacing: touching and regular (solid), touching but random (liquid), far apart (gas).

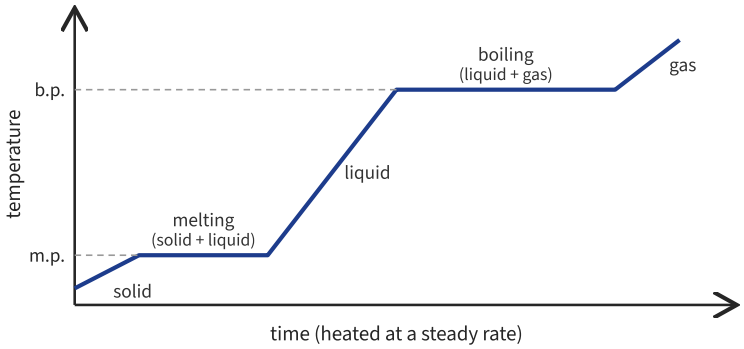
States of matter

	Solid	Liquid	Gas
Shape	fixed	takes the shape of its container	fills its container
Volume	fixed	fixed	no fixed volume
Compress?	no	no (only very slightly)	**yes** — lots of space between particles
Flow?	no	yes	yes
Arrangement	**regular** (lattice)	**random**	**random**
Separation	very close (touching)	close (touching)	**far apart**
Motion	**vibrate** about fixed positions	move around / slide past each other	move **quickly and randomly** in all directions
Forces between particles	strong	weaker	very weak

Changes of state



Change	What happens to the particles
Melting	Heating gives particles more energy so they **vibrate more**. At the melting point they have enough energy to **overcome some of the forces of attraction**, so they can move around each other.
Boiling	At the boiling point particles have enough energy to **overcome the forces of attraction** completely and escape. Bubbles of gas form **throughout** the liquid.
Evaporating	Liquid → gas **below** the boiling point, only at the **surface**: the particles with the most energy escape. Faster at higher temperature and with a larger surface area.
Condensing / freezing	Cooling: particles **lose energy** and slow down, so the forces of attraction pull them closer together. Energy is **given out**.



Heating curve (e.g. ice → water → steam). A cooling curve is the same shape in reverse.

Heating and cooling curves

The temperature rises when one state is being heated. It stays constant while the state is changing (the flat parts).

> ***Why is the line flat? (common 2-mark question)***
> The energy taken in is used to **overcome the forces of attraction** between the particles, not to increase their kinetic energy, so the temperature does not rise.
> Cooling curve: the flat part is where energy is **released** as forces of attraction form, which balances the heat lost.

> ***Which state is it in at a given temperature?***
> Below the melting point → **solid** · between melting and boiling point → **liquid** · above the boiling point → **gas**. e.g. bromine: m.p. −7 °C, b.p. 59 °C. At room temperature (25 °C) it is between the two → **liquid**. Careful with negatives: −10 °C is *below* −7 °C, so at −10 °C bromine is solid.

Pressure and temperature in gases

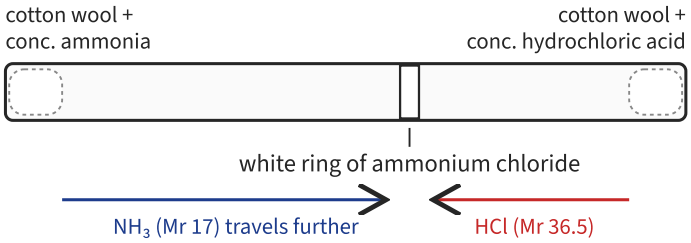
Gas **pressure** is caused by gas particles colliding with the walls of the container.

Change	Effect	Explanation (kinetic particle theory)
Temperature ↑ (pressure kept the same)	volume **increases**	Particles gain kinetic energy and move faster, so they hit the walls **more often and with more force**. The gas expands until the pressure is back to what it was.
Pressure ↑ (temperature kept the same)	volume **decreases**	The particles are pushed **closer together** — gases can be compressed because there is lots of space between the particles.
Temperature ↑ in a sealed, fixed-volume container	pressure **increases**	Particles move faster and collide with the walls **more frequently and with more force**.

Diffusion

> ***Definition***
> **Diffusion** – the net movement of particles from an area of higher concentration to an area of lower concentration, as a result of their random movement.

- Happens in **gases** (fast) and **liquids** (slower). Not in solids — the particles only vibrate in fixed positions.
- Examples: a smell spreading across a room; a purple potassium manganate(VII) crystal dissolving and the colour spreading through water without stirring.
- **Higher temperature → faster diffusion** (particles have more kinetic energy).
- Lower relative molecular mass → faster diffusion: at the same temperature, lighter particles move faster.



$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$. The white ring forms nearer the HCl end because ammonia has the lower M_r, so its molecules move faster and travel further in the same time.

Watch out

✗ "Particles expand when heated" → the particles stay the same size; the **spaces between them** change.
✗ "Particles in a solid don't move" → they **vibrate** about fixed positions.
✗ Diffusion answers without the word **random** movement, or without "high to low concentration".
✗ In the ammonia / HCl experiment, saying the heavier gas moves faster — it's the **lighter** one (NH_3).
✗ Mixing up boiling and evaporating: boiling happens at the boiling point throughout the liquid; evaporation happens below it, at the surface only.

Self-check before the topic questions

- ☐ Describe the properties of solids, liquids and gases (shape, volume, compressibility)
- ☐ Describe and draw particle arrangement, separation and motion in each state
- ☐ Name all the changes of state and explain each using particles and energy
- ☐ Explain the flat sections of heating and cooling curves
- ☐ Work out the state of a substance at a given temperature from its m.p. and b.p.
- ☐ Describe and explain how temperature and pressure affect the volume of a gas
- ☐ Define diffusion and explain the effect of temperature and of M_r on its rate
- ☐ Explain where the white ring forms in the ammonia / hydrogen chloride experiment

Topic 2 Atoms, Elements and Compounds

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–4 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Atomic Structure & the Periodic Table

Elements, compounds and mixtures

	Element	Compound	Mixture
What it is	only **one type of atom**	two or more elements **chemically bonded** together	two or more substances **not chemically combined**
Examples	Fe, He, O_2, Cl_2	H_2O, NaCl, CO_2	air, sea water, crude oil
Proportions	—	**fixed** (shown by the formula)	can vary
Properties	—	**different** from the elements in it	same as the substances in it
Separate by	can't be split into anything simpler	only a **chemical reaction**	**physical methods** (filtering, distillation, chromatography…)

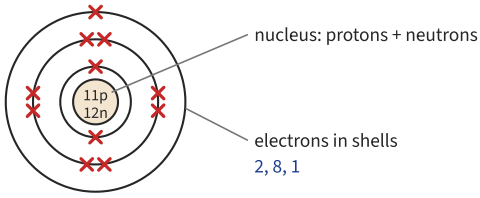
Example that makes it click: iron filings + sulfur powder is a *mixture* — a magnet pulls the iron out. Heat them and they react to make iron(II) sulfide, FeS, a *compound*: it isn't magnetic and you can't get the iron back with a magnet.

> ***Watch out***
> O_2 and Cl_2 are **elements** even though they are molecules — only one type of atom. "Molecule" and "compound" don't mean the same thing.

Atomic structure

An atom has a tiny central **nucleus** containing **protons** and **neutrons**, surrounded by **electrons** in **shells**.

Particle	Relative charge	Relative mass	Where
proton	+1	1	nucleus
neutron	0	1	nucleus
electron	−1	1/1840	shells



Sodium atom, $^{23}_{11}Na$

Atoms are neutral: **number of electrons = number of protons**.

> ***Definitions***
> **Proton number** (atomic number) – the number of protons in the nucleus of an atom.
> **Mass number** (nucleon number) – the total number of protons and neutrons in the nucleus.

Reading an atom symbol

In $^{23}_{11}Na$ the **top** number is the mass number and the **bottom** is the proton number.

protons = **11** · electrons = **11** · neutrons = 23 − 11 = **12**

Neutrons = mass number − proton number

Electronic configuration (elements 1–20)

Shells fill in order: **2** in the 1st, **8** in the 2nd, **8** in the 3rd, then the 4th (K, Ca).

- outer-shell electrons = **group number** (Groups I–VII)
- number of occupied shells = **period number**
- Group VIII (noble gases): **full outer shell**

Atom / ion	H	C	O	Na	Cl	Ar	Ca	Na^+	O^{2-}	Cl^-	Ca^{2+}
Proton number	1	6	8	11	17	18	20	11	8	17	20
Configuration	1	2,4	2,6	2,8,1	2,8,7	2,8,8	2,8,8,2	2,8	2,8	2,8,8	2,8,8

> ***Watch out***
> Work out electrons from the **proton number**, never the mass number. For ions: a **positive** charge means electrons **removed**, a **negative** charge means electrons **added** (O^{2-} has 8 + 2 = 10 electrons).

Isotopes

> ***Definition***
> **Isotopes** – different atoms of the same element that have the same number of protons but different numbers of neutrons.

	$^{35}_{17}Cl$	$^{37}_{17}Cl$	$^{35}_{17}Cl^-$
protons	17	17	17
neutrons	18	20	18
electrons	17	17	18

Isotopes have the same chemical properties because they have the same number of electrons, so the same electronic configuration. (Chemistry is decided by electrons, not neutrons.)

Relative atomic mass, A_r

A_r = the average mass of the isotopes of an element, compared to 1/12 of the mass of a carbon-12 atom. It's an average weighted by how common each isotope is (its **abundance**).

> ***Calculation***
> $A_r = (mass_1 \times \%_1 + mass_2 \times \%_2) \div 100$ e.g. chlorine: 75% Cl-35 and 25% Cl-37
> $A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 =$ **35.5**
> Sense check: the answer must lie between the isotope masses, and nearer the more common one.

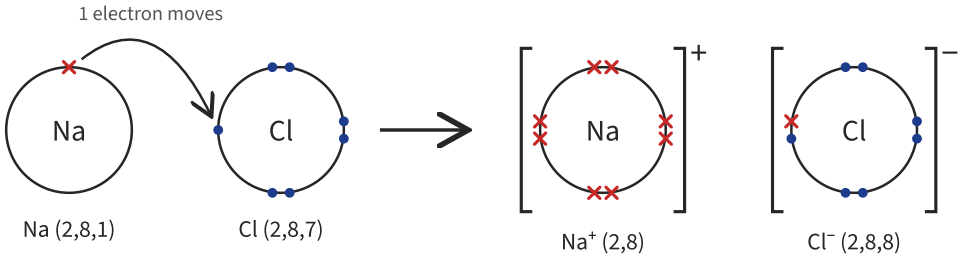
2 Ions & Ionic Bonds

Atoms bond to get a full outer shell. Three ways: **ionic** (metal + non-metal, electrons transferred), **covalent** (non-metals, electrons shared), **metallic** (metal atoms, outer electrons delocalised).

> ***Definitions***
> **Ion** – an atom or group of atoms with a charge, because it has lost or gained electrons.
> **Ionic bond** – the strong electrostatic attraction between oppositely charged ions.

- Metals **lose** electrons → positive ions
- Non-metals **gain** electrons → negative ions
- Charge = electrons lost or gained to reach a full shell

Group	I	II	III	V	VI	VII
Ion charge	+1	+2	+3	−3	−2	−1



Sodium + chlorine → sodium chloride. Na gives its 1 outer electron to Cl; both ions now have a full outer shell.

> ***Drawing ionic dot-and-cross diagrams***
> Dots for one atom's electrons, crosses for the other's · outer shell only unless asked · square brackets round each ion with the charge outside, top right · check every ion has a full outer shell.

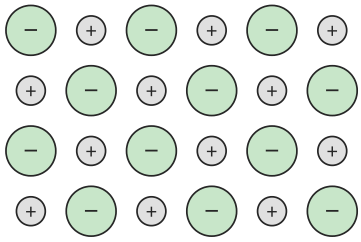
Other examples: Mg (2,8,2) gives 2 electrons to O (2,6) → Mg^{2+} O^{2-} (MgO). Mg gives 1 electron to *each* of two Cl atoms → Mg^{2+} $2Cl^-$ ($MgCl_2$).

Working out ionic formulas

The charges must add up to zero.

$Mg^{2+} + Cl^-$	need 2 Cl^-	→ **$MgCl_2$**
$Al^{3+} + O^{2-}$	2 × (+3) = +6, 3 × (−2) = −6	→ **Al_2O_3**

Shortcut: swap the charge numbers and drop the signs, then simplify (Mg^{2+} O^{2-} → Mg_2O_2 → MgO).



Cl^- (large) Na^+ (small)

NaCl lattice (one layer)

Giant ionic lattice

A regular arrangement of alternating positive and negative ions in 3D. There are no molecules — NaCl just gives the **ratio** of ions (1 : 1).

Property of ionic compounds	Why
High melting and boiling points	Giant lattice with many **strong electrostatic attractions between oppositely charged ions**; a lot of energy is needed to overcome them.
Conduct when molten or in aqueous solution	The **ions are free to move** and carry the charge.
Don't conduct when solid	Ions are held in fixed positions in the lattice and can't move.

Watch out

In ionic compounds it's the **ions** that move, not electrons · NaCl is not a "molecule" and has no "intermolecular forces" · ionic bonds are strong.

3 Simple Molecules & Covalent Bonds

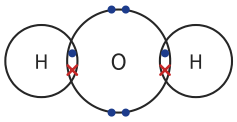
Definition

Covalent bond – a pair of electrons shared between two atoms, giving each atom a noble gas electronic configuration (full outer shell). Forms between non-metals.

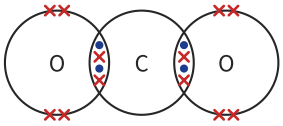
Single bond = 1 shared pair (–) · **double** = 2 pairs (=) · **triple** = 3 pairs (≡).
Electrons not in bonds sit in pairs: **lone pairs**.

Bonds an atom makes = electrons it needs to fill its outer shell:

Atom	H	Cl	O	N	C
Bonds	1	1	2	3	4



H_2O: 2 single bonds, 2 lone pairs on O



CO_2: 2 double bonds

Drawing covalent dot-and-cross diagrams

Overlap the outer shells of bonded atoms · one dot + one cross in the overlap for each bond · leftover electrons in pairs · check every atom has 8 in its outer shell (H has 2), and each atom still shows its own number of outer electrons (O = 6, N = 5, C = 4, H = 1).

Molecules you must be able to draw

Molecule	Displayed	Shared pairs	Lone pairs	Molecule	Displayed	Shared pairs	Lone pairs
H_2	H–H	1	0	**O_2**	O=O	2	2 on each O
Cl_2	Cl–Cl	1	3 on each Cl	**N_2**	N≡N	3	1 on each N
HCl	H–Cl	1	3 on Cl	**CO_2**	O=C=O	4	2 on each O
H_2O	H–O–H	2	2 on O	**C_2H_4**	$H_2C=CH_2$	6	0
NH_3	3 × N–H	3	1 on N	**CH_3OH**	$H_3C–O–H$	5	2 on O
CH_4	4 × C–H	4	0				

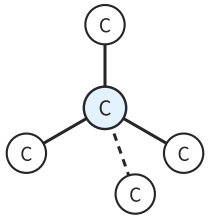
Property of simple molecular substances	Why
Low melting and boiling points (often gases or liquids at room temperature)	Only **weak intermolecular forces** between molecules; little energy is needed to overcome them. The covalent bonds *inside* the molecules are strong and do **not** break.
Poor electrical conductivity	Molecules have no overall charge: **no ions or free electrons** to carry charge.

Watch out — most common lost mark in this topic
"When water boils, the covalent bonds break" is **wrong**: the intermolecular forces are overcome and steam is still H_2O. Covalent bonds are strong; the forces *between* molecules are weak.

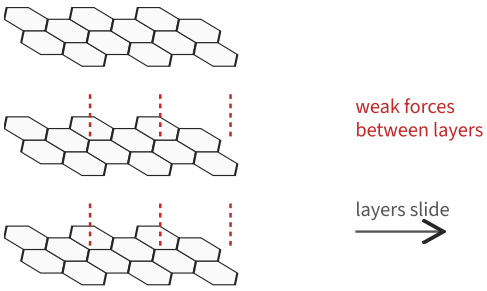
4 Giant Structures

Diamond and graphite

A huge network of atoms all joined by strong covalent bonds — no separate molecules. So: **very high melting points** (many strong covalent bonds to break), insoluble, and they don't conduct — except graphite.



Diamond: each C bonded to 4 others



Graphite: layers of hexagons, each C bonded to 3 others

	Diamond	Graphite
Structure	each C bonded to **4** others; rigid 3D (tetrahedral) network	each C bonded to **3** others in flat layers of hexagons; weak forces between layers
Free electrons	none — all 4 outer electrons used in bonds	1 **delocalised electron** per C atom
Properties	very hard · no conductivity · very high m.p.	soft, slippery · conducts · very high m.p.
Uses	**cutting tools** (drill tips)	**lubricant**, **electrodes**

Link structure to use (common 2–3 mark question)
Diamond → cutting: each C bonded to 4 others by strong covalent bonds in a rigid structure, so it is very hard.
Graphite → lubricant: weak forces between the layers, so the layers slide over each other.
Graphite → electrodes: delocalised electrons can move through the structure, so it conducts; very high melting point too.

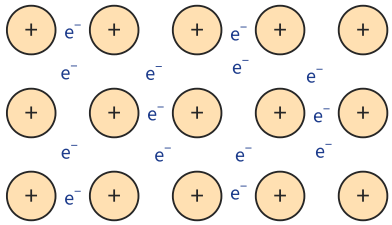
Silicon(IV) oxide

SiO_2 (sand): giant covalent — each **Si bonded to 4 O**, each **O bonded to 2 Si** (hence the 1 : 2 formula). Similar to diamond: hard, very high melting point, doesn't conduct — because both are giant networks of strong covalent bonds with no free electrons.

Metallic bonding

> ***Definition***
> **Metallic bonding** – the electrostatic attraction between the positive ions in a giant metallic lattice and a "sea" of delocalised electrons.

Property	Why
Good electrical conductivity	**Delocalised electrons can move** through the structure and carry charge.
Malleable and ductile	**Layers of positive ions can slide** over each other; the sea of electrons keeps holding them together, so the metal bends instead of breaking.



Positive ions in a sea of delocalised electrons

★ Putting it together

Structure	Examples	Melting point	Conducts as solid?	Conducts molten / dissolved?
Giant ionic	NaCl, MgO	high	no	yes (ions move)
Simple molecular	H_2O, CO_2, CH_4	low	no	no
Giant covalent	diamond, graphite, SiO_2	very high	no (graphite yes)	— (insoluble)
Giant metallic	Cu, Fe, Al	usually high	yes (electrons move)	yes

> ***Identify the structure from data***
> **1.** Conducts as a solid? Yes → metal (or graphite)
> **2.** Conducts only when molten / dissolved? → giant ionic
> **3.** Neither: **low** m.p. → simple molecular; **very high** m.p. → giant covalent

> ***"Explain the melting point" in 3 steps***
> **1.** Name the structure. **2.** Name what must be overcome (attraction between oppositely charged ions / weak intermolecular forces / many strong covalent bonds / attraction between positive ions and delocalised electrons). **3.** "so a lot of / little energy is needed."

Self-check before the topic questions

- ☐ Tell elements, compounds and mixtures apart, with examples
- ☐ Give the relative charge, relative mass and position of protons, neutrons and electrons
- ☐ Work out protons, neutrons and electrons from a symbol like $^{23}_{11}Na$ or $^{35}_{17}Cl^-$
- ☐ Write electronic configurations for elements 1–20 and their ions
- ☐ Define isotopes and explain why they have the same chemical properties
- ☐ Calculate A_r from isotope masses and abundances
- ☐ Draw dot-and-cross diagrams for NaCl, MgO, $MgCl_2$; work out ionic formulas
- ☐ Draw H_2, Cl_2, HCl, H_2O, NH_3, CH_4, O_2, N_2, CO_2, C_2H_4, CH_3OH
- ☐ Explain melting point and conductivity for all four structures
- ☐ Compare diamond, graphite and SiO_2 and link to uses

Topic 3 Stoichiometry

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–2 match the SaveMyExams sub-topics, so you can do their questions after each one · A_r values from the Periodic Table: H 1, C 12, N 14, O 16, Na 23, Mg 24, S 32, Cl 35.5, Ca 40, Fe 56

1 Formulae & Relative Masses

Formulae

> ***Definition***
> **Molecular formula** – the number and type of different atoms in one molecule.

Reading a formula

- H_2SO_4 → 2 H, 1 S, 4 O
- $Ca(OH)_2$ → the 2 multiplies everything in the bracket: 1 Ca, 2 O, 2 H
- $2H_2O$ → the big number multiplies the whole formula: 4 H, 2 O

Elements: these exist as diatomic molecules: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2. Most other elements are just written as the symbol (Fe, C, Na, He).

From a model or diagram: count each type of atom in *one* molecule (2 C, 6 H, 1 O → C_2H_6O). For a giant structure, give the ratio (SiO_2).

Compound	Formula
water	H_2O
carbon dioxide / monoxide	CO_2 / CO
methane	CH_4
ammonia	NH_3
sulfur dioxide	SO_2
hydrochloric acid	HCl
sulfuric acid	H_2SO_4
nitric acid	HNO_3
sodium hydroxide	NaOH
calcium carbonate	$CaCO_3$
ethanol	C_2H_5OH

Empirical formulae & formulae of ionic compounds

> ***Definition***
> **Empirical formula** – the simplest whole number ratio of the different atoms or ions in a compound.

Molecular	C_2H_4	C_2H_6	H_2O_2	$C_6H_{12}O_6$	CH_4
Empirical	CH_2	CH_3	HO	CH_2O	CH_4

Divide every number by the biggest number that goes into all of them. Ionic compounds and giant covalent structures are **always** written as empirical formulae (NaCl, $MgCl_2$, SiO_2).

Ionic formulae from charges — the charges must add up to zero (see Topic 2). Learn these ions:

+1	+2	+3	−1	−2
H^+ Li^+ Na^+ K^+ Ag^+ **NH_4^+**	Mg^{2+} Ca^{2+} Ba^{2+} Zn^{2+} Cu^{2+} Fe^{2+}	Al^{3+} Fe^{3+} Cr^{3+}	Cl^- Br^- I^- **OH^-** **NO_3^-**	O^{2-} S^{2-} **SO_4^{2-}** **CO_3^{2-}**

Bold = compound ions (a group of atoms with one overall charge). Roman numerals give the charge: iron(II) = Fe^{2+}, iron(III) = Fe^{3+}, copper(II) = Cu^{2+}.

Compound	Ions	Balance the charges	Formula
calcium hydroxide	Ca^{2+}, OH^-	1 × (+2), 2 × (−1)	$Ca(OH)_2$
iron(III) oxide	Fe^{3+}, O^{2-}	2 × (+3), 3 × (−2)	Fe_2O_3
copper(II) nitrate	Cu^{2+}, NO_3^-	1 × (+2), 2 × (−1)	$Cu(NO_3)_2$
ammonium sulfate	NH_4^+, SO_4^{2-}	2 × (+1), 1 × (−2)	$(NH_4)_2SO_4$
aluminium sulfate	Al^{3+}, SO_4^{2-}	2 × (+3), 3 × (−2)	$Al_2(SO_4)_3$

Put **brackets** round a compound ion only when you need more than one of it.

Writing equations

1	Write the **word equation**.
2	Write the correct **formula** for every substance.
3	**Balance**: same number of each type of atom on both sides. Only change the **big numbers in front**, never the small ones in a formula.
4	Add **state symbols**.

Symbol	Meaning
(s)	solid
(l)	liquid
(g)	gas
(aq)	aqueous — dissolved in water

Worked example — balancing

methane + oxygen → carbon dioxide + water

$CH_4 + O_2 \rightarrow CO_2 + H_2O$ 4 H on the left, 2 on the right → put 2 in front of H_2O

$CH_4 + O_2 \rightarrow CO_2 + 2H_2O$ now 4 O on the right, 2 on the left → put 2 in front of O_2

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

More balanced examples: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$ · $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$ · $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

Ionic equations

An ionic equation shows only the particles that actually change. **Spectator ions** are in the solution but don't change, so they are left out.

Worked example — hydrochloric acid + sodium hydroxide

1. Full equation with states: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$

2. Split everything that is **(aq) and ionic** into ions: $H^+ + Cl^- + Na^+ + OH^- \rightarrow Na^+ + Cl^- + H_2O$

3. Cross out ions that are the same on both sides (Na^+, Cl^-): **$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$**

Must-know ionic equation	Equation	Reaction
Neutralisation	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	any acid + alkali
Precipitation	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	silver nitrate + a chloride
Displacement	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$	zinc + copper(II) sulfate

Watch out

✗ Changing small numbers to balance ($H_2O \rightarrow H_2O_2$) — that makes a different substance.

✗ Forgetting diatomic elements: oxygen gas is O_2, not O.

✗ Splitting solids, liquids or gases into ions in an ionic equation — only (aq) ionic compounds are split.

✗ Ionic equations must balance **charge** as well as atoms (Zn + Cu^{2+}: +2 on each side).

A_r & M_r

Definitions

Relative atomic mass, A_r – the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of carbon-12.

Relative molecular mass, M_r – the sum of the relative atomic masses of all the atoms in the formula. For ionic compounds it is called relative **formula** mass (still M_r). No units.

Substance	Working	M_r
H_2O	(2 × 1) + 16	18
CO_2	12 + (2 × 16)	44
$CaCO_3$	40 + 12 + (3 × 16)	100
$Ca(OH)_2$	40 + 2 × (16 + 1)	74
$(NH_4)_2SO_4$	2 × (14 + 4) + 32 + (4 × 16)	132

Percentage composition by mass

% of element = (A_r × number of atoms) ÷ M_r × 100

% N in ammonium nitrate, NH_4NO_3 (M_r = 80):

(2 × 14) ÷ 80 × 100 = **35%**

Reacting masses by simple proportion

2.4 g of magnesium makes 4.0 g of magnesium oxide. What mass of oxide does 6.0 g of magnesium make?

Scale up: 6.0 ÷ 2.4 = 2.5 times as much → 4.0 × 2.5 = **10.0 g**

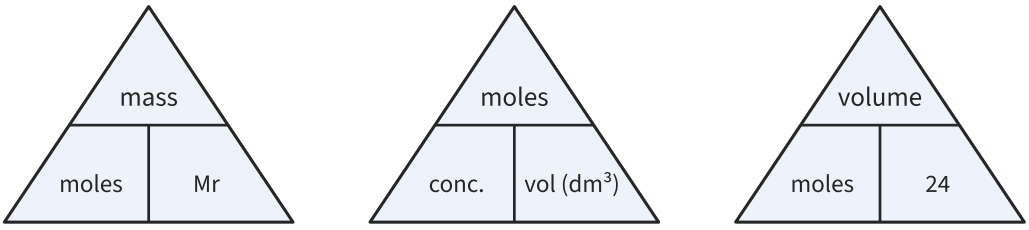
2 The Mole & the Avogadro Constant

The mole

Definition

The **mole** (mol) is the unit of amount of substance. One mole contains 6.02×10^{23} particles (atoms, ions or molecules). This number is the **Avogadro constant**.

- One mole of a substance has a mass equal to its A_r or M_r **in grams** (its molar mass, g/mol): 1 mol C = 12 g; 1 mol H_2O = 18 g; 1 mol CO_2 = 44 g.
- **Number of particles = moles × 6.02×10^{23}**. e.g. 0.5 mol CO_2 = 3.01×10^{23} molecules. Each molecule has 3 atoms, so that is 9.03×10^{23} atoms.



moles = mass (g) ÷ Mr | moles = conc. × volume (dm^3) | gas volume (dm^3) = moles × 24

The three equations for the whole topic. Cover the one you want: side by side means multiply, one above the other means divide.

Linking moles, mass & M_r

Find	Working
Moles in 22 g of CO_2	22 ÷ 44 = **0.5 mol**
Mass of 0.25 mol of $CaCO_3$	0.25 × 100 = **25 g**
M_r of a gas: 0.050 mol has a mass of 3.2 g	3.2 ÷ 0.050 = **64** (matches SO_2: 32 + 32)

Moles and gas volumes

One mole of any gas occupies 24 dm^3 at room temperature and pressure (r.t.p.) — that is 24 000 cm^3. (1 dm^3 = 1000 cm^3.)

Find	Working
Volume of 0.25 mol of O_2	0.25 × 24 = **6.0 dm^3**
Moles in 480 cm^3 of CO_2	480 ÷ 24 000 = **0.020 mol**
Volume of 8.8 g of CO_2	8.8 ÷ 44 = 0.20 mol → 0.20 × 24 = **4.8 dm^3**

For gases, the **volume ratio = the mole ratio** in the equation. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, so 50 cm^3 of methane needs 100 cm^3 of oxygen.

Reacting masses

The 3-step method (works for every reacting-mass question)

1. Moles of the substance you know (mass ÷ M_r). **2.** Use the **balanced equation** to get moles of the one you want. **3.** Convert: mass = moles × M_r, or gas volume = moles × 24.

Worked example 1

Heating 50 g of $CaCO_3$: $CaCO_3 \rightarrow CaO + CO_2$. Mass of CaO? Volume of CO_2?

1. 50 ÷ 100 = 0.50 mol $CaCO_3$

2. ratio 1 : 1 : 1 → 0.50 mol CaO and 0.50 mol CO_2

3. 0.50 × 56 = **28 g CaO** ; 0.50 × 24 = **12 dm³ CO_2**

Worked example 2 — ratio not 1 : 1

Mass of iron from 80 g of Fe_2O_3: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

1. M_r = 160 → 80 ÷ 160 = 0.50 mol

2. 1 Fe_2O_3 : 2 Fe → 1.0 mol Fe

3. 1.0 × 56 = **56 g**

Limiting reactant — the reactant that is used up first. It decides how much product forms; the other reactant is **in excess** (some is left over).

Worked example — limiting reactant

5.6 g of iron is heated with 4.0 g of sulfur: Fe + S → FeS. Which is limiting? Mass of FeS?

Fe: 5.6 ÷ 56 = 0.10 mol · S: 4.0 ÷ 32 = 0.125 mol. The ratio is 1 : 1, so 0.10 mol Fe needs only 0.10 mol S → **iron is limiting** , sulfur is in excess (0.025 mol = 0.8 g left over).

FeS formed = 0.10 mol × 88 = **8.8 g**

Tip: divide each reactant's moles by its number in the equation — the smaller answer is the limiting reactant.

Calculating concentration

Concentration can be measured in **mol/dm³** or **g/dm³**.

> concentration (mol/dm³) = moles ÷ volume (dm³) · concentration (g/dm³) = mass ÷ volume (dm³)
> g/dm³ = mol/dm³ × M_r · **cm³ → dm³: divide by 1000** (250 cm³ = 0.250 dm³)

Find	Working
Conc. of 0.50 mol NaOH in 250 cm³	0.50 ÷ 0.250 = **2.0 mol/dm³** = 2.0 × 40 = **80 g/dm³**
Moles in 25.0 cm³ of 0.100 mol/dm³ HCl	0.100 × 0.0250 = **0.00250 mol**
Mass of NaCl to make 500 cm³ of 0.200 mol/dm³	0.200 × 0.500 = 0.100 mol → × 58.5 = **5.85 g**

Titration calculations

A titration finds the exact volume of one solution that reacts with a known volume of another. Use the **mean of the concordant titres** (results within 0.10 cm³ of each other) and ignore the rough one. e.g. titres 21.00 (rough), 20.40, 20.30, 20.35 → mean = 20.35 cm³.

Worked example 1 (1 : 1)

25.0 cm³ of NaOH is neutralised by 20.0 cm³ of 0.150 mol/dm³ HCl. Concentration of the NaOH? $NaOH + HCl \rightarrow NaCl + H_2O$

1. moles HCl = 0.150 × 0.0200 = 0.00300 mol

2. ratio 1 : 1 → 0.00300 mol NaOH

3. 0.00300 ÷ 0.0250 = **0.120 mol/dm³**

Worked example 2 (2 : 1)

25.0 cm³ of 0.100 mol/dm³ NaOH is neutralised by 12.50 cm³ of H_2SO_4. $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

1. moles NaOH = 0.100 × 0.0250 = 0.00250 mol

2. 2 : 1 → 0.00250 ÷ 2 = 0.00125 mol H_2SO_4

3. 0.00125 ÷ 0.01250 = **0.100 mol/dm³**

Empirical & molecular formula

Worked example — from percentages

A compound is 40.0% C, 6.7% H and 53.3% O by mass. Its M_r is 180. Find the empirical and molecular formulae.

Step	C	H	O
mass (or %)	40.0	6.7	53.3
÷ A_r	40.0 ÷ 12 = 3.33	6.7 ÷ 1 = 6.7	53.3 ÷ 16 = 3.33
÷ smallest (3.33)	1	2.01 ≈ 2	1

Empirical formula = **CH_2O** (mass 12 + 2 + 16 = 30). Molecular: 180 ÷ 30 = 6 → **$C_6H_{12}O_6$**

From experiment masses

2.4 g Mg burns to give 4.0 g of oxide, so oxygen = 4.0 − 2.4 = 1.6 g.
Mg: 2.4 ÷ 24 = 0.10 O: 1.6 ÷ 16 = 0.10 → 1 : 1 → **MgO**

When the ratio ends in .5

Iron oxide: 70% Fe, 30% O. Fe: 70 ÷ 56 = 1.25 O: 30 ÷ 16 = 1.875
÷ 1.25 → 1 : 1.5 → × 2 → 2 : 3 → **Fe_2O_3**

Percentage yield & purity

% yield = actual yield ÷ theoretical yield × 100
% purity = mass of pure product ÷ total mass of sample × 100

Why the yield is below 100%: the reaction may not go to completion (e.g. reversible reactions); some product is lost when transferring, filtering or purifying; other (side) reactions may happen.

Worked example — % yield

From 50 g $CaCO_3$ the theoretical yield of CaO is 28 g (worked example 1 above). Only 22.4 g is made.
22.4 ÷ 28 × 100 = **80%**

Worked example — % purity

12.5 g of impure $CaCO_3$ reacts with excess acid and gives 2.64 dm^3 of CO_2. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
CO_2: 2.64 ÷ 24 = 0.110 mol → 0.110 mol $CaCO_3$ × 100 = 11.0 g
11.0 ÷ 12.5 × 100 = **88%**

Watch out — where calculation marks are lost

✗ Not converting cm^3 to dm^3 (÷ 1000) before using concentration or the 24 dm^3 rule.
✗ Skipping the equation ratio in step 2 — always check the big numbers.
✗ Using A_r of O (16) for oxygen gas: O_2 has M_r 32.
✗ Rounding in the middle of a calculation — keep full numbers, round only the final answer (usually 3 significant figures).
✗ Using the reactant in excess instead of the limiting one to work out the product.
✗ Percentage yield the wrong way up — actual goes on top, so the answer is never over 100%.

Self-check before the topic questions

- ☐ Define molecular formula, empirical formula, A_r, M_r, the mole and the Avogadro constant
- ☐ Write formulae of ionic compounds from the ions, including compound ions and brackets
- ☐ Write and balance symbol equations with state symbols, and write ionic equations
- ☐ Calculate M_r and percentage composition by mass
- ☐ Convert between mass, moles, number of particles and gas volume (24 dm^3 at r.t.p.)
- ☐ Calculate reacting masses and gas volumes, and find the limiting reactant
- ☐ Calculate concentrations in mol/dm^3 and g/dm^3, converting cm^3 to dm^3
- ☐ Use titration results to find a concentration or volume
- ☐ Calculate empirical and molecular formulae from data
- ☐ Calculate percentage yield and percentage purity

Topic 4 Electrochemistry

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–2 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Electrolysis

Electrolysis principles

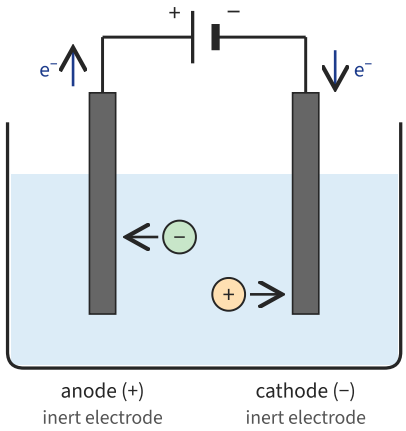
Definitions

Electrolysis – the decomposition of an ionic compound, when molten or in aqueous solution, by the passage of an electric current.

Anode – the **positive** electrode. **Cathode** – the **negative** electrode.

Electrolyte – the molten or aqueous substance that undergoes electrolysis.

- The compound must be **molten or dissolved** so its **ions are free to move**. A solid ionic compound doesn't conduct — its ions are fixed in the lattice.
- **Inert electrodes** (graphite/carbon or platinum) carry the current but don't react.
- Metals or hydrogen form at the **cathode**; non-metals (other than hydrogen) form at the **anode**.
- Memory aid: **PANIC** – **P**ositive is **A**node, **N**egative **I**s **C**athode.



Negative ions move to the anode, positive ions to the cathode. Electrons flow round the wires.

"Describe the transfer of charge during electrolysis"

1. **Electrons** move through the external circuit (wires) from the anode to the cathode.
2. **Ions** move through the electrolyte: positive ions to the cathode, negative ions to the anode.
3. At the **cathode**, positive ions **gain** electrons (reduction). At the **anode**, negative ions **lose** electrons (oxidation).

Oxidation is loss of electrons, reduction is gain (**OIL RIG**). Anode = oxidation, cathode = reduction ("**An Ox**, **Red Cat**").

Electrolysis of molten compounds

Lead(II) bromide, $PbBr_2$, with inert electrodes. Heat the solid until it melts — only then does the bulb light, because the ions can now move. Do it in a fume cupboard: bromine is toxic.

Electrode	Product	Half-equation	What you see
Cathode (−)	lead	$Pb^{2+} + 2e^- \rightarrow Pb$	silvery-grey molten metal forms at the cathode / collects at the bottom
Anode (+)	bromine	$2Br^- \rightarrow Br_2 + 2e^-$	red-brown vapour (fumes) at the anode

Any molten binary compound (two elements): metal at the cathode, non-metal at the anode.

Molten compound	NaCl	KI	$ZnCl_2$	Al_2O_3 (Topic 9)
Cathode	sodium	potassium	zinc	aluminium
Anode	chlorine	iodine	chlorine	oxygen

Electrolysis of aqueous sodium chloride & dilute sulfuric acid

In a solution, water also provides some H^+ and OH^- ions. So two kinds of positive ion compete at the cathode, and two kinds of negative ion at the anode.

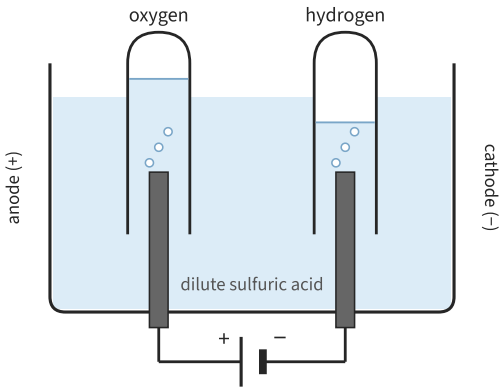
Concentrated aqueous sodium chloride (brine) — ions present: Na^+, H^+, Cl^-, OH^-

Electrode	Product	Half-equation	What you see
Cathode (−)	hydrogen	$2H^+ + 2e^- \rightarrow H_2$	bubbles of colourless gas (sodium is more reactive than hydrogen, so it stays as Na^+)
Anode (+)	chlorine	$2Cl^- \rightarrow Cl_2 + 2e^-$	bubbles of pale yellow-green gas
Left in solution	Na^+ and OH^- → **sodium hydroxide** solution (it becomes alkaline)		

Dilute sulfuric acid — ions present: H^+, SO_4^{2-}, OH^-

Electrode	Product	Half-equation
Cathode (−)	hydrogen	$2H^+ + 2e^- \rightarrow H_2$
Anode (+)	oxygen	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

- Bubbles of colourless gas at both electrodes.
- Twice as much hydrogen as oxygen by volume: the 4 electrons that make 1 O_2 also make 2 H_2.
- Overall, **water** is broken down ($2H_2O \rightarrow 2H_2 + O_2$), so the acid gets more concentrated.



Hydrogen : oxygen = 2 : 1 by volume

Electrolysis of aqueous solutions

Rules for predicting the products

Cathode: the **metal** forms if it is **less reactive than hydrogen** (copper, silver, gold). Otherwise **hydrogen** forms.
Anode: a **halogen** forms from a **concentrated** halide (Cl^-, Br^-, I^-). **Oxygen** forms from a **dilute** halide, or when there is no halide (e.g. sulfate, nitrate).

Reactivity: potassium > sodium > calcium > magnesium > aluminium > zinc > iron > **hydrogen** > copper > silver > gold

Electrolyte (inert electrodes)	Cathode	Anode	Left in solution
Concentrated NaCl(aq)	hydrogen	chlorine	sodium hydroxide
Dilute NaCl(aq)	hydrogen	oxygen	—
Concentrated KBr(aq)	hydrogen	bromine	potassium hydroxide
Concentrated $CuCl_2$(aq)	copper	chlorine	—
Dilute H_2SO_4(aq)	hydrogen	oxygen	sulfuric acid (more concentrated)
$CuSO_4$(aq)	copper	oxygen	sulfuric acid
$AgNO_3$(aq)	silver	oxygen	nitric acid

Aqueous copper(II) sulfate — inert vs copper electrodes

	Carbon (graphite) electrodes	Copper electrodes
Cathode (−)	pink-brown copper coats the electrode $Cu^{2+} + 2e^- \rightarrow Cu$	pink-brown copper coats it; the cathode **gains mass** $Cu^{2+} + 2e^- \rightarrow Cu$
Anode (+)	bubbles of oxygen $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	no gas; the copper anode **dissolves** and **loses mass** $Cu \rightarrow Cu^{2+} + 2e^-$
Solution	blue colour **fades** (Cu^{2+} used up); becomes acidic	blue colour **stays the same** (Cu^{2+} removed at the cathode = Cu^{2+} made at the anode)

The copper-electrode set-up is the idea behind purifying copper: impure copper as the anode, pure copper as the cathode.

Gas	Test
Hydrogen	'pops' with a lighted splint
Oxygen	relights a glowing splint
Chlorine	bleaches damp litmus paper

Ionic half-equations

How to write one

1. Write the ion on the left and the product on the right (halogens, hydrogen and oxygen are **diatomic**: Cl_2, H_2, O_2).
2. Balance the atoms.
3. Add electrons so the **charges balance**: on the **left** at the cathode (gained, reduction), on the **right** at the anode (lost, oxidation).

Cathode — reduction	Anode — oxidation
$Na^+ + e^- \rightarrow Na$	$2Cl^- \rightarrow Cl_2 + 2e^-$
$Ag^+ + e^- \rightarrow Ag$	$2Br^- \rightarrow Br_2 + 2e^-$
$Cu^{2+} + 2e^- \rightarrow Cu$	$2I^- \rightarrow I_2 + 2e^-$
$Pb^{2+} + 2e^- \rightarrow Pb$	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$
$Al^{3+} + 3e^- \rightarrow Al$	$2O^{2-} \rightarrow O_2 + 4e^-$ (molten oxide)
$2H^+ + 2e^- \rightarrow H_2$	$Cu \rightarrow Cu^{2+} + 2e^-$ (copper anode)

Watch out

Write Cl_2, not Cl, at the anode · check the charges: $2Cl^-$ has a total charge of −2, so $2e^-$ must appear on the other side · electrons go on the **left** for the cathode and the **right** for the anode · in solution, sodium, potassium etc. are **never** formed at the cathode — it's hydrogen.

2 Applications of Electrolysis

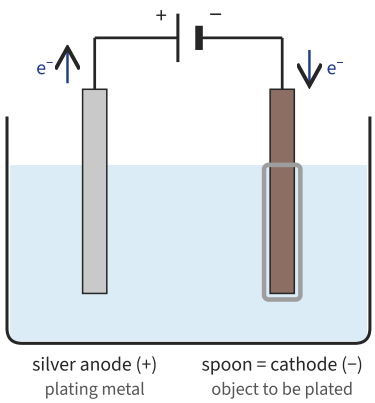
Electroplating

Metal objects are electroplated to improve their appearance and their resistance to corrosion. Examples: chromium-plated steel taps and car parts, silver-plated cutlery, gold-plated jewellery, tin-plated steel cans.

How to electroplate (3 marks)

1. Make the **object the cathode** (−).
2. Make the **plating metal the anode** (+).
3. The **electrolyte** is a solution containing **ions of the plating metal** (e.g. silver nitrate for silver).

Silver plating: $Ag^+ + e^- \rightarrow Ag$ at the cathode (silver coats the spoon); $Ag \rightarrow Ag^+ + e^-$ at the anode (it dissolves), so the concentration of Ag^+ stays the same.



Electrolyte: silver nitrate solution (contains Ag^+ ions)

Hydrogen–oxygen fuel cells

A hydrogen–oxygen fuel cell uses hydrogen and oxygen to produce electricity, with water as the only chemical product: $2H_2 + O_2 \rightarrow 2H_2O$

Advantages vs petrol (gasoline) engines	Disadvantages
• Only product is **water** — no carbon dioxide (greenhouse gas), no toxic carbon monoxide and no oxides of nitrogen from the vehicle • More efficient at producing useful energy • Hydrogen can be made by electrolysis of water (using renewable electricity)	• Hydrogen is a gas: hard to **store** (needs high pressure or large tanks) • Hydrogen is **flammable / explosive** • Few hydrogen filling stations • Most hydrogen is made from fossil fuels, or with electricity from fossil fuels, so CO_2 is still produced somewhere • Fuel cells are expensive

Self-check before the topic questions

- ☐ Define electrolysis, anode, cathode and electrolyte
- ☐ Explain why an ionic compound must be molten or in solution
- ☐ Describe the transfer of charge: electrons in the wires, ions in the electrolyte, gain/loss at the electrodes
- ☐ Give the products and observations for molten $PbBr_2$, concentrated NaCl(aq) and dilute H_2SO_4
- ☐ Predict the products for any molten binary compound and any halide solution (dilute or concentrated)
- ☐ Compare $CuSO_4$(aq) electrolysis with carbon electrodes and with copper electrodes
- ☐ Write half-equations at both electrodes and say which is oxidation and which is reduction
- ☐ Describe how to electroplate an object, and why it is done
- ☐ State what a hydrogen–oxygen fuel cell produces and compare it with a petrol engine

Topic 5 Chemical Energetics

Cambridge IGCSE Chemistry 0620 · Extended · Section 1 matches the SaveMyExams sub-topic, so you can do their questions after it

1 Exothermic & Endothermic Reactions

Endothermic & exothermic reactions

Definitions

Exothermic reaction – transfers thermal energy to the surroundings, leading to an increase in the temperature of the surroundings.

Endothermic reaction – takes in thermal energy from the surroundings, leading to a decrease in the temperature of the surroundings.

	Exothermic	Endothermic
Energy	given **out** to the surroundings	taken **in** from the surroundings
Temperature of surroundings	**rises** — feels warm	**falls** — feels cold
Products vs reactants	products have **less** energy	products have **more** energy
ΔH	**negative**	**positive**
Examples	combustion of fuels, respiration, neutralisation, displacement reactions	thermal decomposition (e.g. $CaCO_3 \rightarrow CaO + CO_2$), photosynthesis, dissolving ammonium nitrate in water
Everyday uses	hand warmers, self-heating cans	instant cold packs for sports injuries

In an experiment, the "surroundings" are the water/solution and the thermometer. Measure the temperature before and after: a rise means exothermic, a fall means endothermic.

Watch out

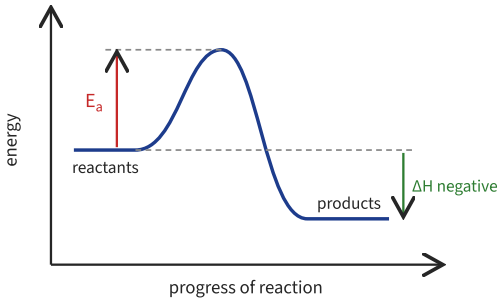
An endothermic reaction gets **colder** because it is taking energy **in** from its surroundings — not because it is "giving out cold".

Enthalpy change & activation energy

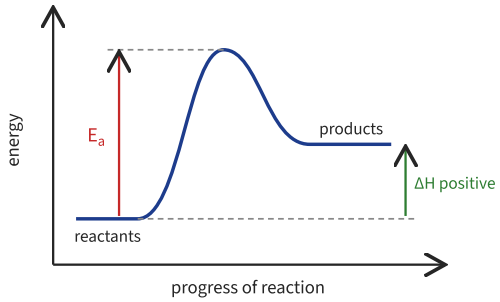
Definitions

Enthalpy change, ΔH – the transfer of thermal energy during a reaction. ΔH is **negative** for exothermic reactions and **positive** for endothermic reactions. Units: kJ/mol.

Activation energy, E_a – the minimum energy that colliding particles must have to react.



Exothermic: products **below** reactants



Endothermic: products **above** reactants

Drawing a reaction pathway diagram

Axes: **energy** (up), **progress of reaction** (across) · reactants and products as labelled horizontal lines (exothermic: products lower; endothermic: products higher), joined by a curve rising to a peak · **E_a**: arrow from the **reactants** level **up** to the peak · **ΔH**: arrow from the **reactants** level to the **products** level — **down** for exothermic, **up** for endothermic.

Watch out

Both arrows start at the **reactants** · the E_a arrow goes to the **peak**, not from the products · an endothermic reaction still has a hump above the products · "activation energy" is a minimum energy for **colliding particles** — use those words.

Bond breaking & bond forming

Key facts

Breaking bonds is endothermic — energy is taken in.

Making bonds is exothermic — energy is released.

Explaining ΔH

Exothermic: more energy is released making the new bonds than is taken in breaking the old bonds.

Endothermic: more energy is taken in breaking bonds than is released making bonds.

ΔH = energy taken in to break bonds − energy released making bonds ("broken minus made"). Bond energies are always given in the question.

Method

1. Write the balanced equation with **displayed formulae** so you can see every bond.
2. Count each type of bond, **multiplied by the number of molecules**.
3. Add up the energy for all bonds **broken** (reactants) and all bonds **made** (products).
4. ΔH = broken − made. Give the sign (+ or −) and units (kJ/mol).

Worked example 1: $H_2 + Cl_2 \rightarrow 2HCl$

H–H = 436, Cl–Cl = 242, H–Cl = 431 kJ/mol

Broken	1 × H–H + 1 × Cl–Cl = 436 + 242	678
Made	2 × H–Cl = 2 × 431	862
ΔH	678 − 862	**−184 kJ/mol**

Negative, so exothermic.

Worked example 2: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

C–H = 413, O=O = 498, C=O = 805, O–H = 464 kJ/mol

Broken	4 × C–H + 2 × O=O = 1652 + 996	2648
Made	2 × C=O + 4 × O–H = 1610 + 1856	3466
ΔH	2648 − 3466	**−818 kJ/mol**

CO_2 is O=C=O (2 double bonds); each H_2O has 2 O–H, so $2H_2O$ has 4.

Try it

$N_2 + 3H_2 \rightarrow 2NH_3$

N≡N = 945, H–H = 436, N–H = 391 kJ/mol

Answer: broken 945 + 3 × 436 = 2253; made 6 × 391 = 2346; ΔH = **−93 kJ/mol**

Try it (endothermic)

$2HBr \rightarrow H_2 + Br_2$

H–Br = 366, H–H = 436, Br–Br = 193 kJ/mol

Answer: broken 2 × 366 = 732; made 436 + 193 = 629; ΔH = **+103 kJ/mol**

Watch out

"Breaking bonds releases energy" is **wrong** · do **broken − made**, not the other way round · multiply by the coefficients in the equation · C=O and C–O have different bond energies — check which one is in the molecule · always give the sign and kJ/mol.

Self-check before the topic questions

- ☐ Define exothermic and endothermic reactions, including what happens to the temperature of the surroundings
- ☐ Give examples of exothermic and endothermic reactions
- ☐ State what ΔH is and its sign for exothermic and endothermic reactions
- ☐ Define activation energy
- ☐ Draw and label reaction pathway diagrams with reactants, products, ΔH and E_a
- ☐ Read a pathway diagram and say whether the reaction is exothermic or endothermic
- ☐ Explain ΔH in terms of bond breaking (endothermic) and bond making (exothermic)
- ☐ Calculate ΔH from bond energies

Topic 6 Chemical Reactions

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–3 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Chemical Change & Rate of Reaction

Physical and chemical changes

	Physical change	Chemical change
New substance?	**No** new substance is made	A **new substance** is made
Reversing it	usually easy (cool it, evaporate the water…)	usually hard to reverse
Examples	melting ice, boiling water, dissolving salt	burning magnesium, rusting, neutralisation
Signs	change of state or appearance only	gas given off (fizzing), colour change, precipitate forms, temperature change

Watch out

Dissolving is a **physical** change: the salt is still there and you get it back by evaporating the water. Always justify your answer with "a new substance is / is not formed".

Rates of reaction factors

The **rate of reaction** is how fast reactants are used up or products are made.

Change	Effect on rate	Notes
Increase concentration of a solution	faster	
Increase pressure of a gas	faster	only matters for reactions involving gases
Increase surface area of a solid	faster	smaller pieces / powder instead of lumps
Increase temperature	faster	has the biggest effect of all the factors
Add a catalyst	faster	e.g. manganese(IV) oxide in $2H_2O_2 \rightarrow 2H_2O + O_2$. **Enzymes** are biological catalysts.

Definition

A **catalyst** increases the rate of a reaction and is unchanged in mass and chemical composition at the end of the reaction.

Collision theory

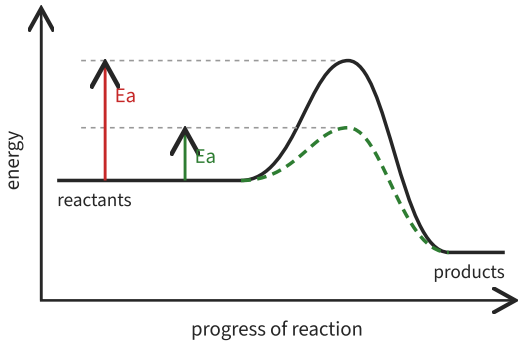
Particles can only react when they **collide**, and the collision must have at least the activation energy. These are **successful collisions**.

Definition

Activation energy, E_a – the minimum energy that colliding particles must have to react.

The rate depends on **how often successful collisions happen**, which depends on:

- the number of particles per unit volume
- the frequency of collisions
- the kinetic energy of the particles
- the activation energy



Solid: no catalyst. Dashed: with a catalyst — an alternative route with a **lower E_a**. The energy of reactants and products doesn't change.

Explaining rates using collision theory

Change	Explanation (mark-scheme wording)
Higher concentration	**More particles per unit volume** → collisions are **more frequent** → more successful collisions per second.
Higher gas pressure	Same particles in a smaller volume: **more particles per unit volume** → more frequent collisions.
Larger surface area	More particles of the solid are **exposed** to the other reactant → more frequent collisions.
Higher temperature	Particles have **more kinetic energy** and move faster, so (1) they collide more often, and (2) **a greater proportion of collisions have energy ≥ E_a**. Reason (2) is the main one.
Catalyst	Provides an alternative pathway with a **lower activation energy** → a greater proportion of collisions are successful. The catalyst is not used up.

Watch out — easy marks lost

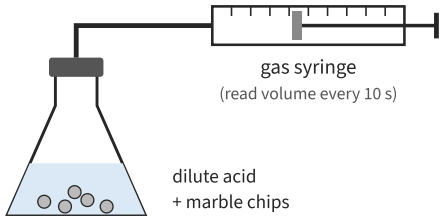
✗ "More collisions" → say **more frequent collisions** or **more collisions per second**.

✗ A catalyst does not give particles more energy — it **lowers E_a**.

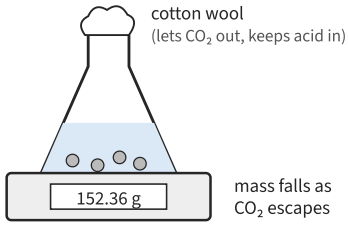
✗ Concentration, pressure and surface area do **not** change E_a or the energy of the particles — only how often they collide.

✗ For temperature, you must mention **energy** (collisions with energy above E_a), not just "particles move faster".

Investigating the rate of a reaction



Method 1: volume of gas produced

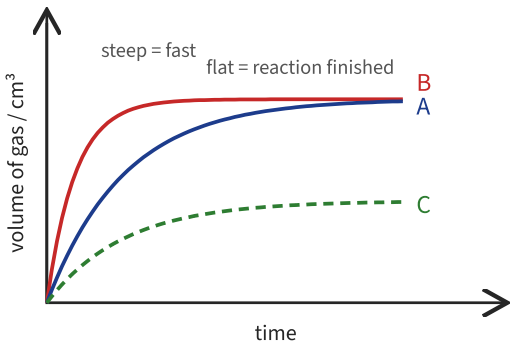


Method 2: loss of mass

Method	How	Evaluation
Volume of gas	e.g. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ or $Mg + 2HCl \rightarrow MgCl_2 + H_2$. Measure the gas volume at regular time intervals.	Gas syringe is accurate. Collecting over water (upturned measuring cylinder) is less accurate — and CO_2 dissolves slightly in water. Some gas escapes before the bung goes in.
Loss of mass	Flask on a balance with a cotton wool plug; record the mass at regular intervals.	Easy and precise for a **dense** gas like CO_2. Useless for H_2 — it is so light the mass change is tiny.
Precipitate ("disappearing cross")	$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$. Time how long until a cross under the flask can't be seen.	Judging when the cross disappears is subjective. Shorter time = faster rate.

For a fair test: change one variable only; keep the others the same (volume and concentration of acid, mass and size of solid, temperature — use a water bath). Start the timer as the reactants are mixed. Repeat and take a mean.

Interpreting data



A original · B faster (e.g. higher temperature or a catalyst), same amounts · C half the amount of the limiting reactant

- The **gradient** (steepness) = the rate.
- Steepest at the **start**: most reactant particles, so collisions are most frequent.
- The gradient falls as reactants are **used up**.
- Line goes **flat**: reaction has stopped because one reactant (the **limiting** reactant) has run out.
- A faster reaction has a **steeper** start and flattens **sooner** — but gives the **same final volume** if the amounts of reactants are the same.
- The final volume only changes if the **amount** of limiting reactant changes.

Calculating rate

Average rate = amount of product formed (or reactant used) ÷ time taken.
e.g. 48 cm^3 of gas in 40 s → 48 ÷ 40 = **1.2 cm^3/s**.
Rate at one moment: draw a tangent to the curve at that time and find its gradient (change in y ÷ change in x).

2 Reversible Reactions & Equilibrium

Reversible reactions

Some reactions can go both ways. The symbol ⇌ shows a **reversible** reaction. Changing the conditions can change the direction.

Hydrated salt		Anhydrous salt + water	Use
$CuSO_4 \cdot 5H_2O$ blue	heat → ← add water	$CuSO_4$ white + $5H_2O$	The forward direction needs heat; adding water to the anhydrous salt reverses it (and gives out heat). Both are used as **tests for water**: white → blue, blue → pink.
$CoCl_2 \cdot 6H_2O$ pink	heat → ← add water	$CoCl_2$ blue + $6H_2O$	

Equilibrium

Definition

A reversible reaction in a **closed system** (nothing can get in or out) is at **equilibrium** when: the rate of the forward reaction equals the rate of the reverse reaction, and the concentrations of reactants and products are no longer changing.

The reactions haven't stopped — both are still happening at the same rate (**dynamic** equilibrium). The **position of equilibrium** tells you whether there is more reactant (left) or more product (right).

Rule: if you change the conditions, the position of equilibrium shifts to oppose the change.

Change	Equilibrium shifts…	Example: $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (forward is exothermic)
Raise temperature	in the **endothermic** direction	left → less NH_3
Lower temperature	in the **exothermic** direction	right → more NH_3
Raise pressure	to the side with **fewer gas molecules**	right (4 molecules → 2) → more NH_3
Lower pressure	to the side with **more gas molecules**	left → less NH_3
Add more reactant	to the **right**, using it up	add N_2 → more NH_3
Remove product	to the **right**, replacing it	remove NH_3 → more NH_3 made
Add a catalyst	**no change** — equilibrium is just reached faster	same yield, sooner

If both sides have the same number of gas molecules, pressure has no effect on the position of equilibrium.

> ***Worked example — "predict and explain"***
> $2NO_2(g)$ brown ⇌ $N_2O_4(g)$ colourless The forward reaction is exothermic.
> **Temperature raised:** equilibrium shifts **left**, in the endothermic direction, to oppose the rise in temperature → more NO_2, so the mixture goes **darker brown**.
> **Pressure raised:** equilibrium shifts **right**, to the side with fewer gas molecules (2 → 1) → more N_2O_4, so the colour gets **paler**.
> Answer pattern: direction of shift + reason (which side / which direction) + effect on the amount or colour.

The Haber process — making ammonia

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \text{forward reaction exothermic}$$

Sources: nitrogen from **air**; hydrogen from **methane** (natural gas). **Conditions:** 450 °C, 20 000 kPa (200 atm), iron catalyst.

Condition	Best for yield	Best for rate	Why the chosen value is used
450 °C	low (forward is exothermic)	high	**Compromise:** a lower temperature would give a higher yield but the rate would be too slow; 450 °C gives a reasonable yield at a reasonable rate.
200 atm	high (4 gas molecules → 2)	high	High pressure increases both yield and rate, but even higher pressures are **too expensive** (strong equipment, energy for compressors) and **too dangerous**.
Iron catalyst	no effect	faster	Equilibrium is reached faster, so more ammonia per hour.

Only a small fraction (about 15%) of the gases react each time. The ammonia is removed by cooling (it liquefies) and the unreacted N_2 and H_2 are recycled.

The Contact process — making sulfur trioxide (for sulfuric acid)

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \text{forward reaction exothermic}$$

Sources: SO_2 from **burning sulfur** ($S + O_2 \rightarrow SO_2$) or **roasting sulfide ores**; oxygen from **air**. **Conditions:** 450 °C, 200 kPa (2 atm), vanadium(V) oxide catalyst (V_2O_5).

Condition	Best for yield	Best for rate	Why the chosen value is used
450 °C	low (forward is exothermic)	high	**Compromise** between a reasonable yield and a reasonable rate.
2 atm	high (3 gas molecules → 2)	high	The yield is **already very high** at low pressure, so the extra cost of high pressure isn't worth it.
V_2O_5 catalyst	no effect	faster	Equilibrium reached faster.

> ***Watch out***
> Know which conditions go with which process: **both 450 °C**, but Haber is **200 atm + iron** and Contact is **2 atm + V_2O_5**. A catalyst never changes the yield.

3 Redox

Oxidation and reduction

> ***Definitions***
> **Oxidation** – gain of oxygen, **loss of electrons**, or an **increase in oxidation number**.
> **Reduction** – loss of oxygen, **gain of electrons**, or a **decrease in oxidation number**.
> **Redox reaction** – a reaction involving simultaneous oxidation and reduction.

OIL RIG: **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain (of electrons).

Reaction	Oxidised	Reduced
$CuO + H_2 \rightarrow Cu + H_2O$	H_2 (gains oxygen)	CuO (loses oxygen)
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$	CO (gains oxygen)	Fe_2O_3 (loses oxygen)
$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$	$Zn \rightarrow Zn^{2+} + 2e^-$ (loses e^-)	$Cu^{2+} + 2e^- \rightarrow Cu$ (gains e^-)
$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$	Br^- (loses e^-)	Cl_2 (gains e^-)

Oxidation numbers

Rules:

- An **uncombined element** is **0** (Fe, O_2, Cl_2).
- A **monatomic ion** = its charge ($Na^+ = +1$, $O^{2-} = -2$).
- In a **compound** they add up to **0**.
- In an **ion** they add up to the **charge**.
- Usually O = −2 and H = +1 in compounds.

Roman numerals in names give the oxidation number: iron(II) = Fe^{2+}, iron(III) = Fe^{3+}, copper(II) = Cu^{2+}, manganate(VII) = Mn at +7.

Worked examples	
S in SO_4^{2-}	$S + 4(-2) = -2 \rightarrow S =$ **+6**
Mn in MnO_4^-	$Mn + 4(-2) = -1 \rightarrow Mn =$ **+7**
Fe in Fe_2O_3	$2Fe + 3(-2) = 0 \rightarrow Fe =$ **+3**
N in NH_4^+	$N + 4(+1) = +1 \rightarrow N =$ **−3**

In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$: Zn goes 0 → +2 (**oxidised**), Cu goes +2 → 0 (**reduced**).

Redox and electron transfer: oxidising and reducing agents

Definitions

Oxidising agent – a substance that oxidises another substance and is itself reduced.
Reducing agent – a substance that reduces another substance and is itself oxidised.

In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$: **Zn is the reducing agent** (it is oxidised); **Cu^{2+} is the oxidising agent** (it is reduced).

Test	Colour change	What it shows
Acidified aqueous potassium manganate(VII)	**purple → colourless**	it is an oxidising agent, so this happens when a **reducing agent** is present
Aqueous potassium iodide	**colourless → brown** (iodine forms)	it is a reducing agent, so this happens when an **oxidising agent** is present

Watch out

The oxidising agent is the one that gets **reduced** (and vice versa) — the most common mix-up. When asked to identify a species, give the actual particle (Cu^{2+}, not "copper").

Self-check before the topic questions

- ☐ Tell physical and chemical changes apart and justify it
- ☐ State the effect of concentration, pressure, surface area, temperature and catalysts on rate
- ☐ Explain each one using collision theory (particles per unit volume, collision frequency, energy, E_a)
- ☐ Describe and evaluate the gas-volume and mass-loss methods
- ☐ Read rate graphs: gradient, why it flattens, same/different final volume; calculate an average rate
- ☐ Describe the reversible reactions of hydrated $CuSO_4$ and $CoCl_2$ with colours
- ☐ Define equilibrium and predict + explain shifts for temperature, pressure, concentration, catalyst
- ☐ State the Haber and Contact equations, sources, conditions, and explain the conditions
- ☐ Define oxidation and reduction (oxygen, electrons, oxidation number) and work out oxidation numbers
- ☐ Identify oxidising and reducing agents; know the $KMnO_4$ and KI colour changes

Topic 7 Acids, Bases and Salts

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–2 match the SaveMyExams sub-topics, so you can do their questions after each one

1 The Characteristic Properties of Acids & Bases

Properties of acids and bases

> ***Definitions***
> **Bases** – oxides or hydroxides of metals (e.g. CuO, MgO, NaOH).
> **Alkalis** – soluble bases (e.g. NaOH, KOH, $Ca(OH)_2$; aqueous ammonia is also an alkali).

Reactions of acids — learn the word patterns

Pattern	Example	What you see
acid + metal → salt + hydrogen	$Mg + 2HCl \rightarrow MgCl_2 + H_2$	fizzing; metal disappears. Only metals above hydrogen in the reactivity series (not Cu, Ag, Au).
acid + base → salt + water	$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$	black solid dissolves, blue solution
acid + alkali → salt + water	$NaOH + HCl \rightarrow NaCl + H_2O$	no visible change; temperature rises
acid + carbonate → salt + water + carbon dioxide	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$	fizzing; the gas turns limewater milky

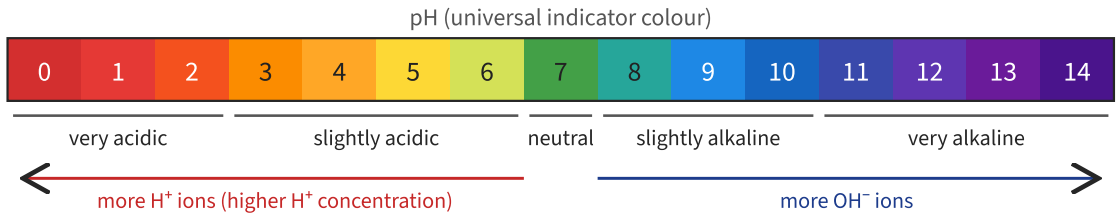
Reactions of bases

Pattern	Example	Notes
base + acid → salt + water	(see above) — this is **neutralisation**	
base + ammonium salt → salt + water + ammonia	$NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$	needs warming; ammonia turns damp red litmus blue

Indicators

Indicator	In acid	In alkali
Litmus	red	blue
Thymolphthalein	colourless	blue
Methyl orange	red	yellow

> ***Watch out***
> Thymolphthalein is **colourless** in acid (not "clear" — clear means see-through, and a coloured solution can be clear).



Universal indicator paper: compare the colour with the chart to find the pH. pH 7 is neutral. The lower the pH, the higher the H^+ ion concentration.

The ions in acids and alkalis

- Aqueous solutions of acids contain H^+ ions.
- Aqueous solutions of alkalis contain OH^- ions.
- **Neutralisation** (acid + alkali) is these ions making water: **$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$**

Proton transfer, strong and weak acids

> ***Definitions***
> **Acid** – a proton donor. **Base** – a proton acceptor. (A proton is an H^+ ion.)
> **Strong acid** – an acid that is completely dissociated in aqueous solution.
> **Weak acid** – an acid that is partially dissociated in aqueous solution.

Example of proton transfer: $HCl + NH_3 \rightarrow NH_4Cl$ — HCl **donates** H^+ (acid), NH_3 **accepts** it (base).

	Strong acid: hydrochloric acid	Weak acid: ethanoic acid
Equation	$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$	$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$
Dissociation	completely (one-way arrow)	partially (reversible arrow — most molecules stay as CH_3COOH)
Same concentration:	more H^+ ions → **lower pH**; reacts **faster** with Mg or carbonates (more vigorous fizzing)	fewer H^+ ions → **higher pH**; reacts **more slowly**

> ***Watch out***
> **Strong ≠ concentrated.** *Strong/weak* is about how completely the acid dissociates. *Concentrated/dilute* is about how much acid is dissolved per dm^3. You can have a dilute strong acid or a concentrated weak acid.

Classifying oxides

Type	Which elements	Examples	Behaviour
Basic	**metal** oxides	CuO, CaO, MgO	react with acids to form a salt + water, e.g. $CaO + 2HCl \rightarrow CaCl_2 + H_2O$. Soluble ones make alkalis ($CaO + H_2O \rightarrow Ca(OH)_2$).
Acidic	**non-metal** oxides	CO_2, SO_2	react with bases / alkalis, e.g. $CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$. Dissolve in water to make acidic solutions.
Amphoteric	a few metals	**Al_2O_3, ZnO**	react with acids *and* with bases to produce a salt and water, e.g. $ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$, and ZnO also reacts with aqueous sodium hydroxide.

The link: metallic character → basic oxide; non-metallic character → acidic oxide. (A few oxides, such as carbon monoxide, are neutral.)

2 Preparation of Salts

A **salt** is made when the H^+ of an acid is replaced by a metal ion (or ammonium ion). The acid decides the second half of the name:

Acid	Hydrochloric acid, HCl	Sulfuric acid, H_2SO_4	Nitric acid, HNO_3
Salt	chloride	sulfate	nitrate

> **Which method?**
> **1.** Is the salt **insoluble**? → **precipitation** (Method C).
> **2.** Soluble, and a **sodium, potassium or ammonium** salt? → **titration** (Method B) — both reactants are soluble, so you can't add an excess and filter it off.
> **3.** Any other soluble salt? → acid + **excess** insoluble base, carbonate or metal, then filter (Method A).

Preparing soluble salts

Method A — excess insoluble solid

e.g. copper(II) sulfate from copper(II) oxide + dilute sulfuric acid

1. Warm the dilute acid (speeds up the reaction).
2. Add the solid a little at a time, stirring, until **no more dissolves** (some is left over — it is in **excess**). This

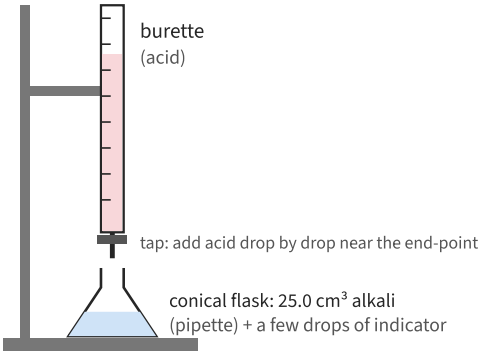
Method B — titration (acid + alkali)

e.g. sodium sulfate: $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

makes sure **all the acid is used up**.

3. **Filter** to remove the excess solid. Keep the filtrate (the salt solution).
4. **Heat** the filtrate to evaporate some of the water until it is **saturated** (crystals form on a glass rod dipped in and taken out).
5. Leave to **cool** so crystals form.
6. Filter off the crystals, wash with a little cold distilled water, and **dry** between filter papers.

With a carbonate (e.g. $ZnCO_3 + H_2SO_4 \rightarrow ZnSO_4 + H_2O + CO_2$) add it until the fizzing stops. With a metal (e.g. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$) use a moderately reactive metal — not sodium or potassium (dangerous) and not copper (doesn't react).



1. Pipette 25.0 cm^3 of alkali into a conical flask; add a few drops of indicator (methyl orange or thymolphthalein).
2. Add acid from the burette until the indicator **just** changes colour (the end-point). Record the volume.
3. **Repeat without the indicator**, adding exactly the same volumes — so the salt isn't contaminated.
4. Evaporate, crystallise and dry as in Method A.

Watch out

Don't evaporate to **dryness** if you want crystals of a hydrated salt — you'd drive off the water of crystallisation (and the solid can spit). Heat only until saturated, then let it cool. Don't use universal indicator in a titration — its colour change is gradual, not sharp.

Preparing insoluble salts — Method C: precipitation

Mix two **solutions**: one containing the metal ion, one containing the non-metal ion. The insoluble salt forms as a **precipitate**.

e.g. barium sulfate:
$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow \mathbf{BaSO_4(s)} + 2NaCl(aq)$
ionic: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

Other examples: $AgNO_3 + NaCl \rightarrow AgCl(s) + NaNO_3$; $Pb(NO_3)_2 + 2KI \rightarrow PbI_2(s) + 2KNO_3$

1. Mix the two solutions; a precipitate forms.
2. **Filter** — the salt is the residue on the filter paper.
3. **Wash** it with distilled water to remove the soluble impurities.
4. **Dry** it (between filter papers or in a warm oven).

Choosing the two solutions

Use a **nitrate** of the metal (all nitrates are soluble) and a **sodium or potassium** compound of the non-metal ion (all are soluble). e.g. for lead(II) iodide: lead(II) nitrate + potassium iodide.

Solubility rules

Soluble	Insoluble
all **sodium, potassium and ammonium** salts	
all **nitrates**	
chlorides …	… except **lead** and **silver** chloride
sulfates …	… except **barium**, **calcium** and **lead** sulfate
… except sodium, potassium and ammonium carbonate	**carbonates**
… except sodium, potassium, ammonium hydroxide (calcium hydroxide is partially soluble)	**hydroxides**

Hydrated and anhydrous salts

Definitions

Hydrated substance – a substance that is chemically combined with water.

Anhydrous substance – a substance containing no water.

Water of crystallisation – the water molecules present in crystals, e.g. $CuSO_4 \cdot 5H_2O$ and $CoCl_2 \cdot 6H_2O$.

Heating removes the water of crystallisation: $CuSO_4 \cdot 5H_2O$ (blue) → $CuSO_4$ (white) + $5H_2O$. Adding water turns it blue again (see Topic 6, reversible reactions).

Worked example — find x in $CuSO_4 \cdot xH_2O$

2.50 g of hydrated copper(II) sulfate is heated until the mass stops changing: 1.60 g of anhydrous $CuSO_4$ is left.

Mass of water lost = 2.50 − 1.60 = 0.90 g

Moles of $CuSO_4$ = 1.60 ÷ 160 = 0.010 mol (M_r = 64 + 32 + 4 × 16 = 160)

Moles of H_2O = 0.90 ÷ 18 = 0.050 mol

Ratio $CuSO_4 : H_2O$ = 0.010 : 0.050 = 1 : 5 → **x = 5**, so $CuSO_4 \cdot 5H_2O$

Heat to **constant mass** so you know all the water has gone.

Self-check before the topic questions

- ☐ Give the word and symbol equations for acids with metals, bases, alkalis and carbonates
- ☐ Describe the reaction of a base with an ammonium salt and the test for ammonia
- ☐ State the colours of litmus, thymolphthalein and methyl orange in acid and alkali
- ☐ Use universal indicator and the pH scale to compare acidity and H^+ concentration
- ☐ Write the ionic equation for neutralisation
- ☐ Define acids and bases as proton donors/acceptors and strong/weak acids; write the HCl and ethanoic acid equations
- ☐ Classify oxides as acidic, basic or amphoteric (Al_2O_3, ZnO)
- ☐ Choose and describe the right method to make a salt (excess solid, titration, precipitation)
- ☐ Use the solubility rules
- ☐ Define hydrated, anhydrous and water of crystallisation; work out x in a hydrated formula

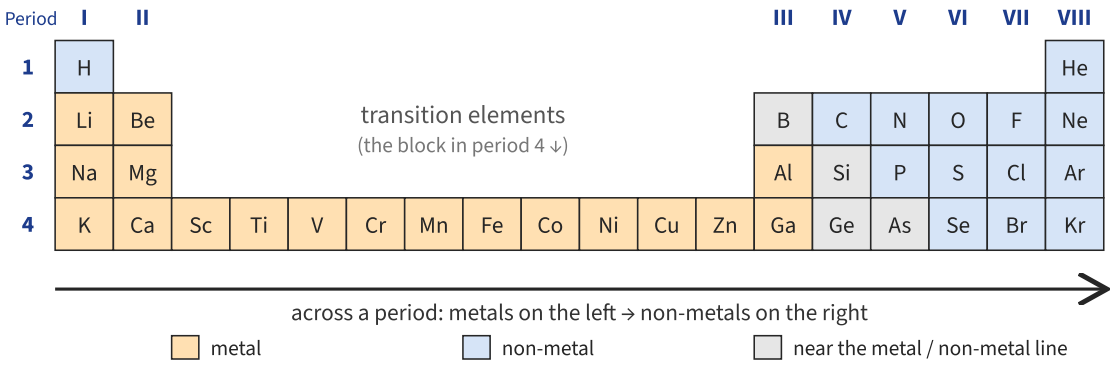
Topic 8 The Periodic Table

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–2 match the SaveMyExams sub-topics, so you can do their questions after each one

1 The Periodic Table & Trends

The Periodic Table

> ***Definition***
> The Periodic Table is an arrangement of elements in order of increasing proton number (atomic number), in **periods** and **groups**.
> **Group** = a vertical column · **Period** = a horizontal row



First four periods. Groups are numbered I–VIII; the transition elements sit between Groups II and III.

Position	What it tells you	Example: Mg
Group number (I–VII)	number of electrons in the **outer shell**	Group II → 2 outer electrons
Period number	number of **occupied electron shells**	Period 3 → 3 shells
Group VIII	noble gases — **full outer shell**	so Mg = **2,8,2**

Periodic Trends

Across a period the elements change from metallic to non-metallic character: metals on the left (form positive ions), non-metals on the right (form negative ions or share electrons).

Down a group the elements have similar chemical properties because they have the **same number of electrons in their outer shell**.

Group	I	II	III	IV	V	VI	VII	VIII
Outer e⁻	1	2	3	4	5	6	7	8*
Ion charge	+1	+2	+3	—	−3	−2	−1	—

*helium has 2. Groups I–III lose electrons; V–VII gain them. Group IV and the noble gases don't usually form ions.

> ***Exam wording***
> "Why do sodium and potassium react in a similar way?" → **They both have one electron in their outer shell** (same electronic configuration in the outer shell). Mention the *outer shell* — "same group" alone doesn't score.

2 Group Properties & Trends

Group I Properties — the alkali metals

Lithium, sodium and potassium are relatively soft metals (they can be cut with a knife). They react with air and water, so they are stored under oil.

Trend down the group	Li → Na → K
Melting point	**decreases** (Li 181 °C, Na 98 °C, K 63 °C)
Density	generally **increases**
Reactivity	**increases**

> ***Why reactivity increases down Group I***
> The outer electron is **further from the nucleus** and more **shielded** by inner shells, so it is less strongly attracted and is **lost more easily**.

Reaction with cold water: metal + water → metal hydroxide + hydrogen. The solution formed is **alkaline** (hence "alkali metals").

$2Na + 2H_2O \rightarrow 2NaOH + H_2$ (same pattern for Li and K: $2Li \rightarrow 2LiOH$, $2K \rightarrow 2KOH$)

Metal	What you see with water
Lithium	floats, fizzes steadily, slowly gets smaller
Sodium	floats, melts into a ball, fizzes quickly, moves around on the surface
Potassium	as sodium but more violent; the hydrogen ignites with a **lilac flame**

> ***Predicting the next element (common question)***
> Use the trends: **rubidium** (below K) will have a **lower melting point** than potassium, a **higher density**, and react **even more violently** with water to form rubidium hydroxide (RbOH) and hydrogen. Always use the data you are given and quote the trend.

Group VII Properties — the halogens

Halogens are diatomic non-metals: Cl_2, Br_2, I_2 — two atoms joined by a covalent bond.

Halogen	Appearance at r.t.p.
Chlorine, Cl_2	**pale yellow-green gas**
Bromine, Br_2	**red-brown liquid**
Iodine, I_2	**grey-black solid**

Learn these three exactly — the colour words are what score.

Trend down the group	Cl → Br → I
Density	**increases**
Reactivity	**decreases**
Colour	gets darker
State	gas → liquid → solid (melting and boiling points increase)

> ***Why reactivity decreases down Group VII***
> Halogens react by **gaining** an electron. Further down, the outer shell is **further from the nucleus** and more shielded, so an extra electron is **attracted less strongly** and gained less easily. (The opposite of Group I, which reacts by *losing* an electron.)

Group VII Displacement Reactions

A more reactive halogen displaces a less reactive halogen from an aqueous solution of its halide ions.

Halogen added ↓ / solution →	potassium chloride (Cl^-)	potassium bromide (Br^-)	potassium iodide (I^-)
Chlorine	—	turns **orange** (Br_2 formed)	turns **brown** (I_2 formed)
Bromine	no reaction	—	turns **brown** (I_2 formed)
Iodine	no reaction	no reaction	—

Symbol equations

$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$

$Cl_2 + 2KI \rightarrow 2KCl + I_2$

$Br_2 + 2KI \rightarrow 2KBr + I_2$

Ionic equations (K^+ is a spectator ion)

$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$

$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$

$Br_2 + 2I^- \rightarrow 2Br^- + I_2$

These are redox reactions

The halogen **gains electrons** → it is **reduced** (it is the oxidising agent): $Cl_2 + 2e^- \rightarrow 2Cl^-$

The halide ion **loses electrons** → it is **oxidised**: $2Br^- \rightarrow Br_2 + 2e^-$

Observation answer: "the colourless solution turns orange / brown" — give the colour *change*, not just the final colour.

Transition Elements

The transition elements are metals that have high densities, high melting points, form coloured compounds, and often act as catalysts (as the element or in compounds).

	Transition elements	Group I metals
Density	high	low
Melting point	high	low
Hardness	hard, strong	soft
Compounds	**coloured**	white / colourless solutions
Catalysts?	**often**	no

Coloured compounds: copper(II) sulfate solution is blue; iron(II) compounds are usually pale green; iron(III) compounds are usually orange-brown.

Catalysts: iron in the Haber process; vanadium(V) oxide in the Contact process.

Variable oxidation numbers

Transition elements can form ions with different oxidation numbers. The Roman numeral in the name gives it: **iron(II)** = Fe^{2+}, e.g. $FeCl_2$, FeO · **iron(III)** = Fe^{3+}, e.g. $FeCl_3$, Fe_2O_3

Noble Gases

Definition

The Group VIII noble gases are unreactive, monatomic gases. This is because they have a **full outer shell** of electrons (a stable electronic configuration), so they don't need to lose, gain or share electrons.

Monatomic = they exist as single atoms (He, Ne, Ar), not molecules like Cl_2. Because they are unreactive, argon is used to fill light bulbs and helium to fill balloons.

Watch out

- "Noble gases have 8 outer electrons" — helium has 2. Say **full outer shell**.
- Chlorine vs chlorid**e**: chlorine is the element (Cl_2); chloride is the ion (Cl^-). Write Cl_2, not Cl, for the halogen in equations.
- Displacement only goes one way: a **less** reactive halogen can't displace a more reactive one (iodine + potassium chloride = no reaction).
- Reactivity goes **up** down Group I but **down** down Group VII — don't mix them up.
- "Group I reactivity increases because there are more electrons" — no: it's because the outer electron is **further from the nucleus** and more **shielded**.

Self-check before the topic questions

☐ Describe how the Periodic Table is arranged (groups, periods, proton number)

☐ Link group number to outer electrons and ion charge; period number to shells

☐ Describe metallic → non-metallic character across a period

☐ Explain why elements in the same group react similarly

☐ Give the Group I trends and predict the properties of Rb or Cs

☐ Write the equation and observations for Li, Na and K with water

☐ State the appearance of Cl_2, Br_2, I_2 and the Group VII trends

☐ Explain halogen displacement: observations, equations, what is oxidised / reduced

☐ State the four properties of transition elements; explain iron(II) / iron(III)

☐ Explain why the noble gases are unreactive and monatomic

Topic 9 Metals

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–3 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Properties, Uses & Alloys of Metals

Properties of Metals

Physical property	Metals	Non-metals
Thermal conductivity	**good**	poor
Electrical conductivity	**good**	poor (except graphite)
Malleable and ductile?	**yes** — can be hammered into shape and drawn into wires	no — brittle when solid
Melting and boiling points	usually **high**	usually low (except e.g. diamond, graphite)

Chemical properties — how metals react:

With	General equation	Example
Oxygen	metal + oxygen → metal oxide	$2Mg + O_2 \rightarrow 2MgO$ (bright white flame) $2Cu + O_2 \rightarrow 2CuO$ (black solid forms)
Cold water	metal + water → metal hydroxide + hydrogen	$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$
Steam	metal + steam → metal oxide + hydrogen	$Mg + H_2O \rightarrow MgO + H_2$
Dilute acid	metal + acid → salt + hydrogen	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Uses of Metals

A use is always explained by a **property** — learn them as pairs:

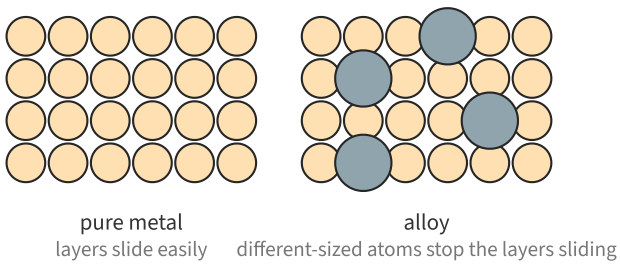
Use	Property that makes it suitable
Aluminium — aircraft	**low density**
Aluminium — overhead electrical cables	**low density** and **good electrical conductivity**
Aluminium — food containers	**resistance to corrosion**
Copper — electrical wiring	**good electrical conductivity** and **ductile**

Alloys

> ***Definition***
> **Alloy** – a mixture of a metal with other elements.

Alloy	Made of
Brass	copper and zinc
Stainless steel	iron with other elements such as **chromium, nickel and carbon**

- Alloys can be harder and stronger than the pure metals, so they are often more useful.
- **Stainless steel → cutlery**, because it is **hard** and **resistant to rusting**.



pure metal
layers slide easily

alloy
different-sized atoms stop the layers sliding

> ***Why alloys are harder (2–3 marks)***
> In a pure metal the atoms are all the same size and arranged in regular layers that can slide over each other. In an alloy, the **different sized atoms** disrupt the layers, so the **layers can no longer slide over each other** easily.
> **Spotting an alloy in a diagram:** look for atoms of a *different size* mixed in among the metal atoms.

2 Reactivity Series & Corrosion of Metals

Reactivity Series

Most reactive → least reactive (carbon and hydrogen are included as reference points):

K **Na** **Ca** **Mg** **Al** **C** **Zn** **Fe** **H** **Cu** **Ag** **Au**

Memory aid: **P**lease **S**end **C**ats, **M**onkeys **A**nd **C**ute **Z**ebras **I**nto **H**ot **C**ountries **S**igned **G**ordon (potassium, sodium, calcium, magnesium, aluminium, carbon, zinc, iron, hydrogen, copper, silver, gold).

Metal	Cold water	Steam	Dilute hydrochloric acid
Potassium	very vigorous; H_2 ignites (lilac flame)	—	dangerously violent — never done
Sodium	vigorous; melts into a ball, fizzes		
Calcium	steady fizzing; solution goes cloudy		vigorous fizzing
Magnesium	very slow	**reacts**: bright white glow → MgO + H_2	fizzes quickly
Zinc	(not needed for the syllabus)		fizzes steadily
Iron			fizzes slowly → iron(II) chloride
Copper, silver, gold	**no reaction**		**no reaction**

$Mg + 2HCl \rightarrow MgCl_2 + H_2$ · $Fe + 2HCl \rightarrow FeCl_2 + H_2$ · $2K + 2H_2O \rightarrow 2KOH + H_2$

> ***Explaining the acid reactions***
> Metals **above hydrogen** in the reactivity series react with dilute acids, displacing hydrogen. Metals **below hydrogen** (copper, silver, gold) do not react with dilute acids.

> ***Deducing an order from results (common question)***
> Rank the metals by how vigorously they react: more bubbles / faster fizzing / bigger temperature rise / more displacement reactions = **more reactive**. Example: X fizzes rapidly in acid, Y fizzes slowly, Z does nothing → order **X > Y > Z**, and Z is below hydrogen.

Explaining Reactivity

The reactivity of a metal is its tendency to form positive ions (to lose electrons). The more easily a metal loses electrons, the more reactive it is.

Displacement: a more reactive metal displaces a less reactive metal from an aqueous solution of its ions.

Metal added ↓ / solution →	$MgSO_4$	$ZnSO_4$	$FeSO_4$	$CuSO_4$
Magnesium	—	✓	✓	✓
Zinc	✗	—	✓	✓
Iron	✗	✗	—	✓
Copper	✗	✗	✗	—

$Mg + CuSO_4 \rightarrow MgSO_4 + Cu$

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$

Ionic: $Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu$

Ionic: $Cu + 2Ag^{+} \rightarrow Cu^{2+} + 2Ag$

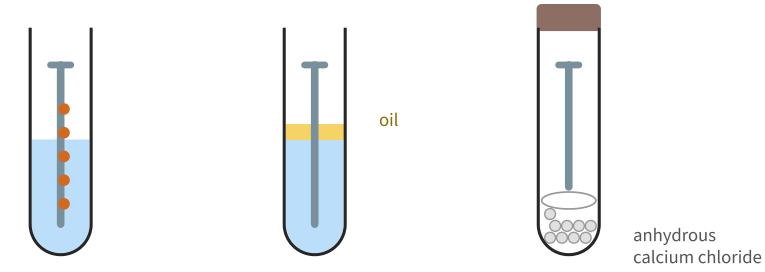
The metal atom loses electrons (is **oxidised**); the metal ion gains them (is **reduced**).

What you see with $CuSO_4$: the blue solution fades (turns pale green with iron) and a pink-brown coating of copper forms on the metal.

> ***Why aluminium seems unreactive***
> Aluminium is high in the reactivity series, but it is covered by a thin **layer of aluminium oxide** that stops water and acids reaching the metal underneath.

Rusting of Iron

> ***Definition***
> Iron and steel rust when they are in contact with both water and oxygen. Rust is **hydrated iron(III) oxide**.
> iron + oxygen + water → hydrated iron(III) oxide



A: air + water
rusts

B: boiled water + oil layer
no rust (no oxygen)

C: dry air, drying agent
no rust (no water)

Only tube A has both water and oxygen, so only A rusts. Boiling removes dissolved oxygen; the oil stops air dissolving back in.

Barrier methods stop rusting by excluding oxygen or water: **painting** (cars, bridges), **greasing / oiling** (moving parts, tools), **coating with plastic** (garden furniture, wire fences).

Galvanising & Sacrificial Protection

> ***Galvanising***
> Coating iron or steel with a layer of **zinc**. It works in two ways:
> **1. Barrier:** the zinc keeps out oxygen and water.
> **2. Sacrificial protection:** even if the zinc is scratched, the iron still doesn't rust.

> ***Explaining sacrificial protection***
> Zinc is **more reactive than iron**, so zinc **loses electrons / is oxidised in preference to iron**:
> $Zn \rightarrow Zn^{2+} + 2e^-$
> The iron doesn't lose electrons, so it doesn't rust.
> Blocks of zinc (or magnesium) attached to ships' hulls work in the same way.

3 Extraction of Metals

Extraction of Metals

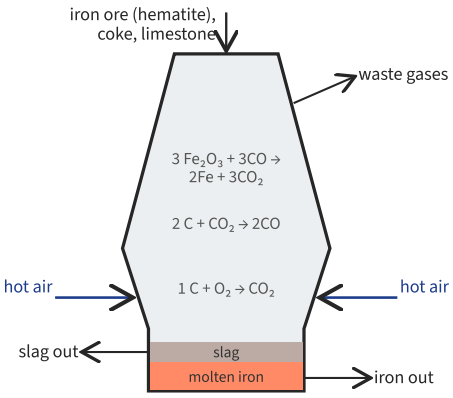
The more reactive the metal, the harder it is to extract from its ore, because it holds on to the other elements in its compounds more strongly.

Metals	Position	How they are extracted
K, Na, Ca, Mg, Al	above carbon	**electrolysis** of the molten compound
Zn, Fe, Cu	below carbon	**reduction by heating with carbon** (carbon takes the oxygen)
Au (and some Ag)	very unreactive	found **uncombined** as the element

e.g. $2CuO + C \rightarrow 2Cu + CO_2$ (copper(II) oxide is reduced; carbon is oxidised)

Extraction of Iron from Hematite

#	Equation	What happens
1	$C + O_2 \rightarrow CO_2$	coke **burns** in hot air, giving heat (exothermic) and carbon dioxide
2	$C + CO_2 \rightarrow 2CO$	carbon dioxide is **reduced** to carbon monoxide by more coke
3	$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$	iron(III) oxide is **reduced** to iron by carbon monoxide (the reducing agent)
4	$CaCO_3 \rightarrow CaO + CO_2$	**thermal decomposition** of limestone to calcium oxide
5	$CaO + SiO_2 \rightarrow CaSiO_3$	calcium oxide (basic) reacts with silicon dioxide (acidic) to form **slag**, calcium silicate



The blast furnace

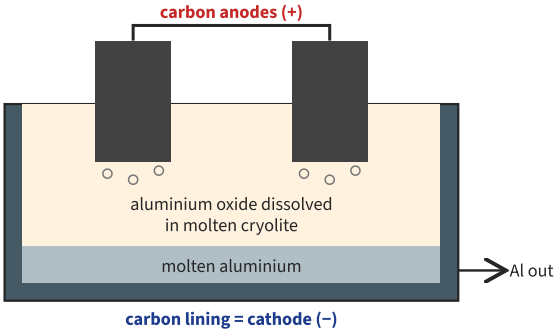
Raw material	Job
Hematite	iron ore: iron(III) oxide, Fe_2O_3
Coke (carbon)	burns to give **heat**; makes the reducing agent CO
Limestone	$CaCO_3$; removes the acidic impurity **silicon dioxide (sand)** as slag
Hot air	provides oxygen for the coke to burn

Molten iron sinks to the bottom; the slag is less dense and floats on top of it. Both are tapped off separately.

Extraction of Aluminium from Bauxite

The main ore of aluminium is **bauxite**. Aluminium is above carbon, so it is extracted by electrolysis of purified bauxite (aluminium oxide, Al_2O_3).

- **Cryolite:** aluminium oxide is dissolved in molten cryolite. This **lowers the melting point**, so less energy is needed (cheaper).
- **Electrodes:** both are carbon (graphite).
- **Anodes replaced regularly:** the oxygen made at the anode reacts with the carbon anode to form carbon dioxide, so the anodes **burn away**: $C + O_2 \rightarrow CO_2$



Electrode	Half-equation	Type
Cathode (−)	$Al^{3+} + 3e^- \rightarrow Al$	reduction — molten aluminium collects at the bottom
Anode (+)	$2O^{2-} \rightarrow O_2 + 4e^-$	oxidation — oxygen gas forms

Watch out

- Rusting needs **both** water and oxygen. Only iron and steel "rust"; other metals *corrode*.
- Rust is **hydrated** iron(III) oxide — not just "iron oxide".
- Galvanised iron still works when scratched because of **sacrificial protection**, not because of the barrier.
- In the blast furnace, **carbon monoxide** reduces the iron(III) oxide. Limestone is there to **remove impurities** as slag, not to make iron.
- Cryolite **lowers the melting point** — it is not a catalyst. Anodes are replaced because they **react with oxygen**, not because they melt.
- Aluminium is a **reactive** metal; it only seems unreactive because of its oxide layer.
- Uses questions: always give the **property** (e.g. "low density"), not just "it is light" or "it is strong".

Self-check before the topic questions

- ☐ Compare metals and non-metals; describe reactions with O_2, water / steam, acid
- ☐ Link the uses of aluminium and copper to their properties
- ☐ Define an alloy (brass, stainless steel); explain why alloys are harder
- ☐ Recite the reactivity series (with C and H)
- ☐ Describe K, Na, Ca + water; Mg + steam; metals + dilute HCl
- ☐ Deduce a reactivity order from experimental results
- ☐ Explain reactivity as forming positive ions; write displacement equations
- ☐ Explain why aluminium seems unreactive
- ☐ State the conditions for rusting and three barrier methods
- ☐ Explain galvanising and sacrificial protection
- ☐ Write all five blast furnace equations and the job of each raw material
- ☐ Explain the aluminium cell: cryolite, anodes, both half-equations

Topic 10 Chemistry of the Environment

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1 Water & Water Pollution

Water: chemical tests

Test for water	Colour change if water is present
Anhydrous copper(II) sulfate	**white → blue**
Anhydrous cobalt(II) chloride	**blue → pink**

The copper(II) sulfate change is reversible:
$CuSO_4 + 5H_2O \rightleftharpoons CuSO_4 \cdot 5H_2O$ (anhydrous, white ⇒ hydrated, blue).

Testing purity: pure water melts at exactly 0 °C and boils at exactly 100 °C. Impurities lower the melting point, raise the boiling point, and make it melt or boil over a **range** of temperatures.

Why distilled water is used in practical chemistry: it contains **fewer chemical impurities** than tap water, so dissolved substances in tap water can't react with or interfere with your results.

> ***Watch out***
> Copper(II) sulfate and cobalt(II) chloride only show that water is **present**. They do not show the water is **pure** — for purity, use the melting and boiling points.

Substances in water from natural sources

Substance	Good or bad?	Why
Dissolved oxygen	beneficial	Needed by aquatic life (e.g. fish) to respire.
Metal compounds	both	Some provide **essential minerals** for life; some are **toxic**.
Plastics	harmful	Harm aquatic life.
Sewage	harmful	Contains **harmful microbes** which cause disease.
Nitrates (from fertilisers)	harmful	Lead to **deoxygenation** of water and damage to aquatic life.
Phosphates (from fertilisers and detergents)	harmful	

> ***How nitrates and phosphates remove oxygen***
> They make algae grow very fast (an algal bloom) → the algae block light, so plants underneath die → bacteria decompose the dead material and **use up the dissolved oxygen** → fish and other aquatic life die.

Water treatment (domestic water supply)

Sedimentation — water stands still; large solid particles settle out → **Filtration** — water passes through sand / gravel; removes smaller solids → **Carbon** — removes tastes and odours → **Chlorination** — chlorine kills microbes

> ***Exam wording — one job per step***
> Sedimentation and filtration **remove solids** · carbon **removes tastes and odours** · chlorination **kills microbes**. Don't say chlorine "cleans" the water or that filtration removes bacteria.

Fertilisers

- Ammonium salts and nitrates are used as fertilisers.
- **NPK fertilisers** provide the elements **nitrogen (N), phosphorus (P) and potassium (K)** for improved plant growth.
- Ammonium salts are made by reacting ammonia with an acid:
  $NH_3 + HNO_3 \rightarrow NH_4NO_3$
  $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

Fertiliser	Formula	Supplies
ammonium nitrate	NH_4NO_3	N
ammonium sulfate	$(NH_4)_2SO_4$	N
ammonium phosphate	$(NH_4)_3PO_4$	N and P
potassium nitrate	KNO_3	K and N

Read the formula to see which of N, P, K it supplies.

2 Air Quality & Climate

Air

Clean, dry air is approximately 78% nitrogen, 21% oxygen, and the remainder is a mixture of **noble gases** (mainly argon) and **carbon dioxide**.



↑ about 1%: noble gases + carbon dioxide

Air pollutants — source and effect

Pollutant	Source	Adverse effect
Carbon dioxide, CO_2	**Complete** combustion of carbon-containing fuels	Higher levels → increased **global warming**, which leads to **climate change**
Carbon monoxide, CO	**Incomplete** combustion of carbon-containing fuels	**Toxic** gas
Particulates (soot)	**Incomplete** combustion of carbon-containing fuels	Increased risk of **respiratory problems and cancer**
Methane, CH_4	**Decomposition of vegetation** and **waste gases from digestion in animals**	Higher levels → increased **global warming** → climate change
Oxides of nitrogen, NO_x	**Car engines**	**Acid rain**, **photochemical smog** and **respiratory problems**
Sulfur dioxide, SO_2	Combustion of **fossil fuels which contain sulfur compounds**	**Acid rain**

How oxides of nitrogen form in car engines

The very high temperature inside the engine makes **nitrogen and oxygen from the air** react:
$N_2 + O_2 \rightarrow 2NO$ then $2NO + O_2 \rightarrow 2NO_2$

Acid rain

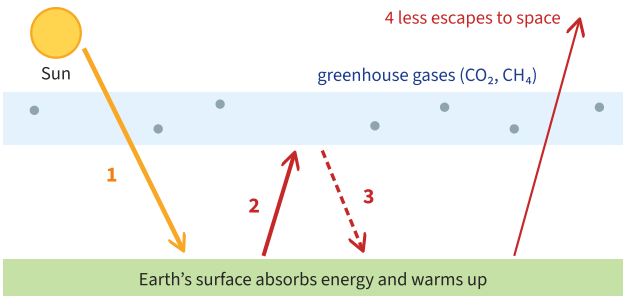
SO_2 and NO_2 dissolve in rainwater to form acids, which damage limestone buildings and harm trees and aquatic life.

Watch out — don't mix them up

CO_2 and CH_4 → global warming / climate change · **SO_2 and NO_x** → acid rain · **CO** → toxic · complete combustion gives CO_2; **incomplete** gives CO and particulates.

Effects of greenhouse gases

1. Thermal energy from the Sun is **absorbed** by the Earth's surface, which warms up.
2. The surface **emits thermal energy** (infrared radiation).
3. CO_2 and CH_4 **absorb** it and **re-emit** it in all directions, some back to Earth (some is also reflected back).
4. This reduces thermal energy loss to space → **global warming** → **climate change**.



Watch out

Say greenhouse gases **absorb and re-emit thermal energy** and **reduce heat loss to space**. "They trap the Sun's rays" is too vague, and the ozone layer has nothing to do with it.

Reducing the effects of environmental issues

Problem	Strategy	Why it works
Climate change	Planting trees	Trees remove CO_2 from the air by photosynthesis.
	Reduction in livestock farming	Fewer animals → less methane from digestion.
	Decreasing use of fossil fuels	Less CO_2 released by combustion.
	Increasing use of hydrogen and renewable energy (e.g. wind, solar)	They release no CO_2 when used — hydrogen burns to form only water.
Acid rain	Catalytic converters in vehicles	Convert oxides of nitrogen into nitrogen (see below).
	Using low-sulfur fuels	Less SO_2 is formed when the fuel burns.
	Flue gas desulfurisation with calcium oxide	Calcium oxide (a base) removes SO_2 from power station waste gases: $CaO + SO_2 \rightarrow CaSO_3$

Catalytic converters

Inside the converter, a catalyst (e.g. platinum or rhodium) turns **carbon monoxide and nitrogen monoxide** into **carbon dioxide and nitrogen**: $2CO + 2NO \rightarrow 2CO_2 + N_2$

So toxic CO and acid-rain-causing NO are removed together. (The N_2 goes back into the air, where it came from.)

Photosynthesis

Definition

Photosynthesis – the reaction between carbon dioxide and water to produce glucose and oxygen, in the presence of chlorophyll and using energy from light.

Word equation	carbon dioxide + water → glucose + oxygen
Symbol equation	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Link: photosynthesis removes CO_2 from the air — that is why planting trees helps against climate change.

Self-check before the topic questions

- ☐ Describe the tests for water (copper(II) sulfate, cobalt(II) chloride) and how to test its purity
- ☐ Explain why distilled water is used in practical chemistry
- ☐ List substances found in natural water and say which are beneficial and which are harmful
- ☐ Explain how nitrates and phosphates lead to deoxygenation
- ☐ Describe the four stages of water treatment and what each one does
- ☐ State that ammonium salts and nitrates are fertilisers; say which of N, P, K a fertiliser supplies
- ☐ State the composition of clean, dry air
- ☐ Give the source and effect of CO_2, CO, particulates, CH_4, NO_x and SO_2
- ☐ Explain how greenhouse gases cause global warming
- ☐ Explain how NO_x forms in engines and write the catalytic converter equation
- ☐ Give strategies to reduce climate change and acid rain, and explain each one
- ☐ Describe photosynthesis and write its word and symbol equations

Topic 11 Organic Chemistry

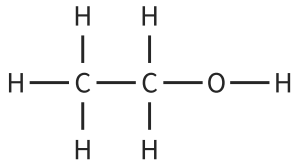
Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–3 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Formulae, Functional Groups & Terminology

Organic formulae

A **hydrocarbon** is a compound that contains hydrogen and carbon only. Ethanol (C_2H_5OH) is *not* a hydrocarbon — it contains oxygen.

Formula type	What it shows	Ethanol
Molecular	number of each type of atom	C_2H_6O
General	formula for a whole series	$C_nH_{2n+1}OH$
Structural	an **unambiguous** description of how the atoms are arranged	CH_3CH_2OH
Displayed	**all** the atoms and **all** the bonds	see right →



ethanol, displayed

More structural formulae: $CH_2{=}CH_2$ (ethene), CH_3COOCH_3 (methyl ethanoate).

Homologous series

> ***Definitions***
> **Homologous series** – a family of similar compounds with similar chemical properties due to the presence of the same functional group.
> **Functional group** – an atom or group of atoms that determine the chemical properties of a homologous series.

General characteristics of a homologous series: same functional group · same general formula · each member differs from the next by a **–CH_2–** unit · a **trend in physical properties** (e.g. boiling point rises as the chain gets longer) · **similar chemical properties**.

Series	Functional group	General formula	Name ends	Example
Alkanes	C–C single bonds only	C_nH_{2n+2}	-ane	methane CH_4
Alkenes	C=C	C_nH_{2n}	-ene	ethene C_2H_4
Alcohols	–OH	$C_nH_{2n+1}OH$	-ol	ethanol C_2H_5OH
Carboxylic acids	–COOH	$C_nH_{2n+1}COOH$	-oic acid	ethanoic acid CH_3COOH
Esters	–COO–	—	-yl …-oate	ethyl ethanoate $CH_3COOC_2H_5$

Saturated & unsaturated compounds

> ***Definitions***
> **Saturated** – molecules in which all carbon–carbon bonds are single bonds (e.g. alkanes).
> **Unsaturated** – molecules in which one or more carbon–carbon bonds are not single bonds (e.g. alkenes, with C=C).

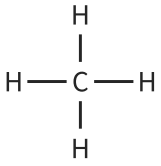
Test: **aqueous bromine** (bromine water) turns from **orange to colourless** with an unsaturated compound; it stays orange with a saturated one (see Alkenes).

Naming organic compounds

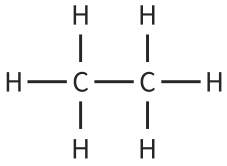
Carbons	Prefix
1	meth-
2	eth-
3	prop-
4	but-

- **Name = prefix (number of C atoms in the chain) + ending (the series).**
- Count **every** carbon, including the one in –COOH: ethan**oic** acid, CH_3COOH, has 2.
- A number shows **where** the C=C or –OH is: count from the end of the chain nearest to it. But-**1**-ene has C=C starting at carbon 1; propan-**2**-ol has –OH on carbon 2.

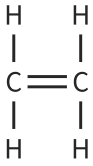
C	Alkane	Alkene	Alcohol	Carboxylic acid
1	methane CH_4	—	methanol CH_3OH	methanoic acid HCOOH
2	ethane CH_3CH_3	ethene $CH_2{=}CH_2$	ethanol CH_3CH_2OH	ethanoic acid CH_3COOH
3	propane $CH_3CH_2CH_3$	propene $CH_2{=}CHCH_3$	propan-1-ol $CH_3CH_2CH_2OH$ propan-2-ol $CH_3CH(OH)CH_3$	propanoic acid CH_3CH_2COOH
4	butane $CH_3CH_2CH_2CH_3$	but-1-ene $CH_2{=}CHCH_2CH_3$ but-2-ene $CH_3CH{=}CHCH_3$	butan-1-ol $CH_3CH_2CH_2CH_2OH$ butan-2-ol $CH_3CH(OH)CH_2CH_3$	butanoic acid $CH_3CH_2CH_2COOH$



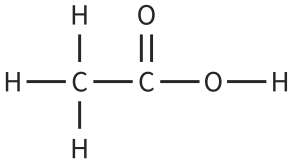
methane



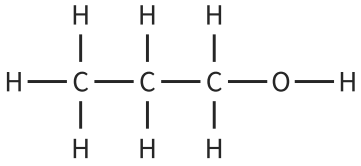
ethane



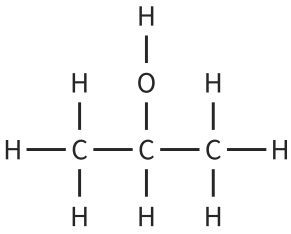
ethene



ethanoic acid



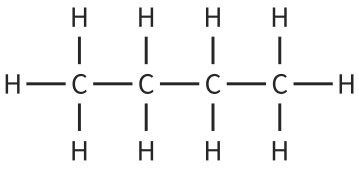
propan-1-ol



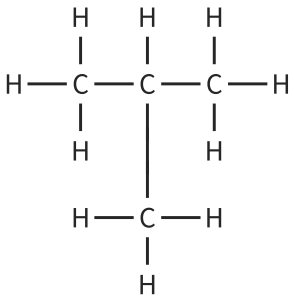
propan-2-ol

Structural isomers

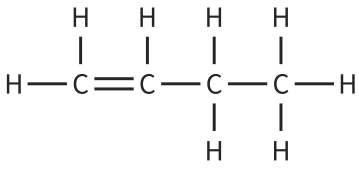
> ***Definition***
> **Structural isomers** – compounds with the same molecular formula but different structural formulae.



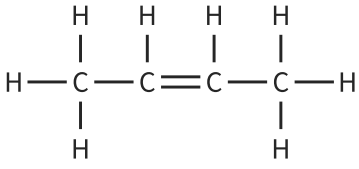
butane, $CH_3CH_2CH_2CH_3$



methylpropane, $CH_3CH(CH_3)CH_3$



but-1-ene, $CH_3CH_2CH{=}CH_2$



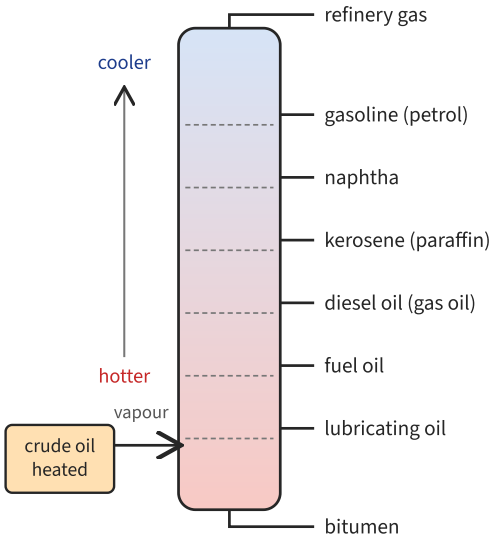
but-2-ene, $CH_3CH{=}CHCH_3$

Both pairs: same molecular formula (C_4H_{10}; C_4H_8), different arrangement. Propan-1-ol and propan-2-ol (above) are isomers too.

2 Organic Families

Fossil fuels

- The fossil fuels are **coal, natural gas and petroleum** (crude oil).
- **Methane** is the main constituent of natural gas.
- Petroleum is a **mixture of hydrocarbons**, separated into useful **fractions** by fractional distillation.



1. Petroleum is heated so it turns into vapour, which enters the column near the bottom.
2. The column is **hot at the bottom and cooler at the top**.
3. Vapours rise. Each fraction **condenses** at the height where the temperature falls below its boiling point, and is collected there.
4. Small molecules (low boiling points) reach the top; bitumen (very high boiling point) is collected at the bottom.

From the bottom to the top of the column
chain length **decreases** · volatility **increases** · boiling point **decreases** · viscosity **decreases**

Fraction (top → bottom)	Use
refinery gas	gas for heating and cooking
gasoline / petrol	fuel for cars
naphtha	chemical feedstock (raw material for making other chemicals)
kerosene / paraffin	jet fuel
diesel oil / gas oil	fuel for diesel engines
fuel oil	fuel for ships and home heating systems
lubricating oil	lubricants, waxes and polishes
bitumen	making roads

Order follows the Cambridge syllabus list. Some textbooks swap fuel oil and lubricating oil — check your teacher's diagram.

Alkanes

Alkanes are saturated hydrocarbons: all their bonds are **single covalent bonds**. They are **generally unreactive**, except for **combustion** and **substitution by chlorine**.

Combustion

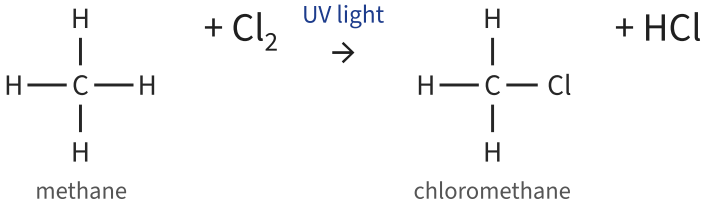
complete: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

incomplete: $2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$

(very little oxygen can also give soot: $CH_4 + O_2 \rightarrow C + 2H_2O$)

Substitution
Substitution reaction – one atom or group of atoms is replaced by another atom or group of atoms.

Substitution with chlorine is a **photochemical reaction**: ultraviolet light provides the activation energy. One H is replaced by Cl:



$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$ (chloroethane)

Alkenes

Alkenes contain a **C=C double covalent bond**, so they are unsaturated hydrocarbons.

Cracking

Alkenes (and hydrogen) are made by **cracking**: breaking **larger alkane molecules** into smaller ones using a high temperature (550 °C) and a catalyst.

$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$ (alkane → smaller alkane + alkene)

$C_2H_6 \rightarrow C_2H_4 + H_2$ (alkane → alkene + hydrogen)

Why crack? (common 2–3 mark question)

1. Fractional distillation gives more large alkanes than are needed, and not enough of the smaller, more useful ones such as petrol — cracking **matches supply to demand**.
2. It produces **alkenes**, used to make polymers and ethanol.
3. It produces **hydrogen**, which can be used as a fuel.

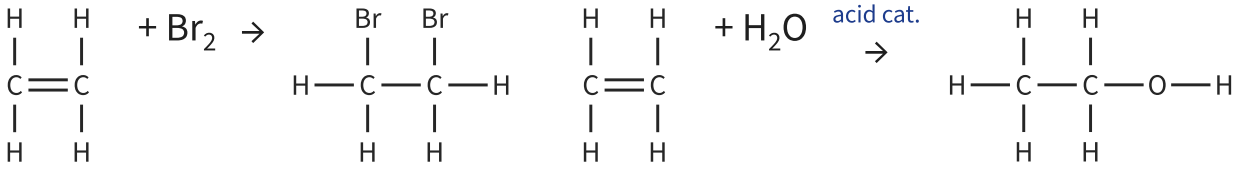
Test: saturated or unsaturated?

Add **aqueous bromine** (bromine water) and shake. Alkene (unsaturated): **orange → colourless**. Alkane (saturated): **stays orange**.

Addition reactions

In an **addition reaction** only one product is formed. The C=C opens up and the atoms add across it.

Add	Conditions	Equation (with ethene)	Product
Bromine	room temperature (bromine or aqueous bromine)	$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$	1,2-dibromoethane
Hydrogen	**nickel catalyst**, heated	$C_2H_4 + H_2 \rightarrow C_2H_6$	ethane
Steam (H_2O)	**acid catalyst** (phosphoric acid), 300 °C, 60 atm (6000 kPa)	$C_2H_4 + H_2O \rightarrow C_2H_5OH$	ethanol



ethene | 1,2-dibromoethane | ethene | ethanol

Other alkenes do the same: propene + Br_2 → 1,2-dibromopropane, $CH_3CHBrCH_2Br$.

Alcohols

Ethanol is used as a **solvent** and as a **fuel**. It burns completely to carbon dioxide and water:

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

	Fermentation	Catalytic addition of steam to ethene
Equation	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$	$C_2H_4 + H_2O \rightarrow C_2H_5OH$
Conditions	aqueous glucose, **yeast**, **25–35 °C**, **no oxygen**	**300 °C**, **6000 kPa / 60 atm**, **acid catalyst**
Advantages	uses **renewable** raw materials (sugar from crops); low temperature, so less energy needed	**fast**, **continuous** process; produces **pure** ethanol; high yield
Disadvantages	**slow**, batch process; ethanol is **impure** (dilute, must be separated by fractional distillation); needs land for crops	ethene comes from petroleum — **non-renewable**; high temperature and pressure need a lot of energy

Why no oxygen in fermentation? With oxygen, the ethanol is oxidised to ethanoic acid. Why 25–35 °C? Too cold is too slow; too hot denatures the enzymes in the yeast.

Carboxylic acids

Ethanoic acid is a **weak acid**. It reacts like other acids, forming salts called **ethanoates**:

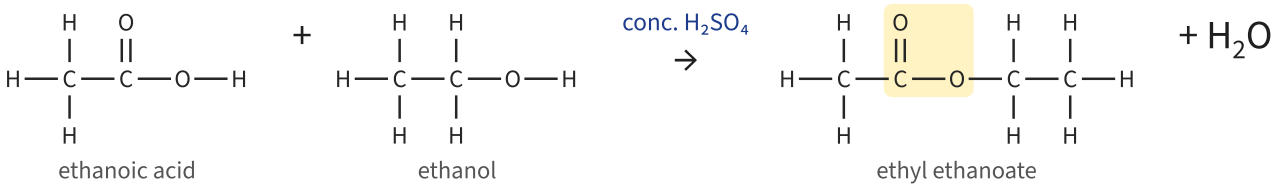
With	Equation	Salt formed
metal	$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$	magnesium ethanoate
base	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	sodium ethanoate
carbonate	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$	sodium ethanoate

Making ethanoic acid by oxidising ethanol:

- with **acidified aqueous potassium manganate(VII)** (purple → colourless): $C_2H_5OH + 2[O] \rightarrow CH_3COOH + H_2O$
- by **bacterial oxidation** during vinegar production (oxygen from the air): $C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$

Ethanoic acid & esterification reactions

A carboxylic acid reacts with an alcohol, using an **acid catalyst** (concentrated sulfuric acid) and heat, to form an ester and water:



ethanoic acid + ethanol → ethyl ethanoate + H_2O

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

Naming esters

Alcohol part first (-ol → **-yl**), then **acid part** (-oic acid → **-oate**).

ethan**ol** + ethan**oic acid** → eth**yl** ethan**oate**.

But the formula is written **acid part first**: $CH_3COO–C_2H_5$.

Alcohol + acid	Ester
methanol + ethanoic acid	methyl ethanoate CH_3COOCH_3
ethanol + methanoic acid	ethyl methanoate $HCOOC_2H_5$
methanol + propanoic acid	methyl propanoate $CH_3CH_2COOCH_3$
propan-1-ol + ethanoic acid	propyl ethanoate $CH_3COOCH_2CH_2CH_3$
butan-1-ol + ethanoic acid	butyl ethanoate $CH_3COOCH_2CH_2CH_2CH_3$

3 Polymers

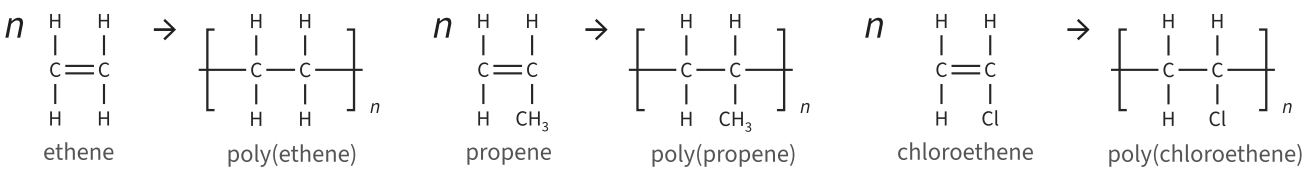
Polymers

Definition

Polymers – large molecules built up from many smaller molecules called monomers. Plastics are made from polymers.

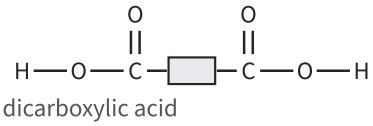
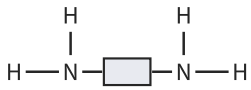
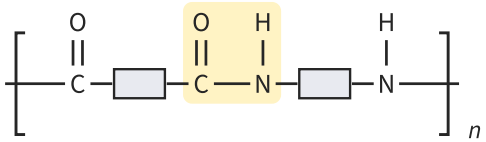
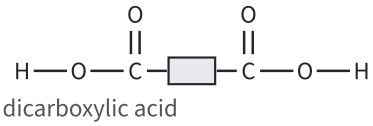
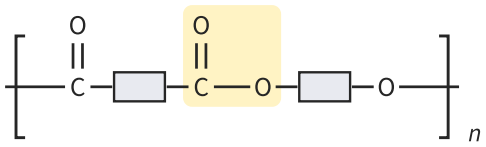
Addition & condensation polymers

Addition polymerisation: monomers with a **C=C** join together; each double bond opens up and the polymer is the **only product**.



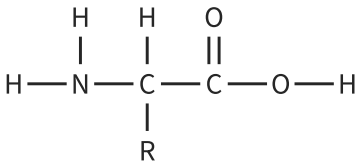
> ***Deducing repeat unit ↔ monomer***
> **Monomer → repeat unit:** draw the two C=C carbons with their groups, change C=C to C–C, add a bond sticking out on each side through brackets, and an *n*.
> **Repeat unit → monomer:** remove the brackets, *n* and the outer bonds, and put back the C=C between the two carbons.

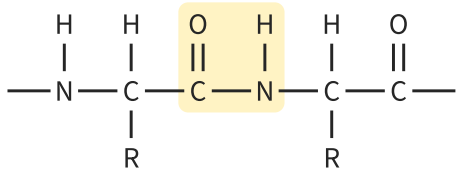
Condensation polymerisation: two different monomers, each with **two functional groups**, join together and a small molecule (water) is formed at every link. The boxes below stand for chains of carbon atoms.

Polymer	Monomers	Repeat unit
Nylon **a polyamide**	 dicarboxylic acid  diamine	 **amide linkage** –CONH– (shaded)
PET **a polyester**	 dicarboxylic acid  diol	 **ester linkage** –COO– (shaded)

PET can be **converted back into its monomers** and re-polymerised, so it can be recycled.

Proteins are natural polyamides, formed from **amino acid** monomers. They have the same amide linkage as nylon; R is a different group in each amino acid.


amino acid, $H_2N–CHR–COOH$


part of a protein chain (amide linkage shaded)

	Addition polymerisation	Condensation polymerisation
Monomers	contain a C=C double bond	each has two functional groups (–COOH, $–NH_2$, –OH); usually two different monomers
Products	**polymer only**	polymer **+ a small molecule (water)**
Linkage	C–C chain	amide –CONH– or ester –COO–
Examples	poly(ethene), poly(propene), poly(chloroethene)	nylon, PET, proteins

Plastics & their disposal

Problem	Why
Disposal in landfill sites	Most plastics are **non-biodegradable**, so they stay in landfill for a very long time and take up land.
Accumulation in oceans	Plastics don't break down, so they build up in the oceans and harm aquatic life.
Burning	Forms **toxic gases** (and adds CO_2, a greenhouse gas).

Watch out — marks often lost in this topic

- Bromine water goes **colourless**, not "clear" (clear can still be coloured).
- Displayed formulae must show **every** bond, including O–H; every carbon has exactly 4 bonds.
- Cracking needs a **catalyst and a high temperature**. It breaks molecules (chemical change); fractional distillation only separates them (physical).
- Chlorine + alkane needs **UV light** and is **substitution** (2 products). Bromine + alkene is **addition** (1 product).
- Ester names: alcohol part first (ethyl), acid part second (ethanoate).
- Addition polymerisation: **no** water formed. Condensation: water formed at every link.
- Fermentation must be done **without oxygen**.

Self-check before the topic questions

- ☐ Draw displayed formulae and write structural, molecular and general formulae
- ☐ Define homologous series, functional group, saturated, unsaturated, structural isomers
- ☐ Name and draw unbranched alkanes, alkenes, alcohols and carboxylic acids up to 4 carbons
- ☐ Name and draw esters made from these alcohols and acids
- ☐ Describe fractional distillation, the trends up the column and the use of each fraction
- ☐ Write combustion and substitution (UV) equations for alkanes
- ☐ Describe cracking (conditions, equations, reasons) and the bromine water test
- ☐ Describe addition of bromine, hydrogen (Ni) and steam (acid catalyst) to alkenes
- ☐ Compare fermentation and the hydration of ethene, with conditions, pros and cons
- ☐ Write reactions of ethanoic acid with metals, bases and carbonates; describe its formation by oxidation
- ☐ Describe esterification and draw an ester
- ☐ Deduce addition polymer repeat units and monomers; draw nylon, PET and a protein
- ☐ Compare addition and condensation polymerisation; explain the problems of plastic disposal

Topic 12 Experimental Techniques and Chemical Analysis

Cambridge IGCSE Chemistry 0620 · Extended · Sections 1–3 match the SaveMyExams sub-topics, so you can do their questions after each one

1 Experimental Techniques

Measuring apparatus

Measure	Apparatus	Notes
Time	stopwatch / stop-clock	units: seconds (s) or minutes
Temperature	thermometer	units: °C
Mass	(top-pan) balance	units: g
Volume (cm^3)	**measuring cylinder**	quick, but the **least precise**; fine when an approximate volume is enough
	volumetric pipette	delivers **one fixed volume** (e.g. 25.0 cm^3) very precisely; use a pipette filler, never your mouth
	burette	delivers **variable** volumes precisely (0–50 cm^3); used in titrations
	gas syringe	measures the volume of a gas (usually up to 100 cm^3)

Advantages and disadvantages — what examiners look for

Pipette or burette vs measuring cylinder: pipette/burette are more precise; measuring cylinder is quicker but less precise.

Gas syringe vs collecting gas over water: the syringe gives a more precise reading and works for gases that dissolve in water (e.g. NH_3, HCl, SO_2); collecting over water can't be used for soluble gases.

Beakers only give a rough volume — never choose one to *measure* volume.

Acid–base titration

Used to find the exact volume of one solution that reacts with a known volume of another (e.g. to find a concentration, or to make a pure salt).

1. Rinse the burette with the acid, then fill it. Remove the air bubble from the tip and record the **initial reading** (read the bottom of the meniscus at eye level).
2. Use a **volumetric pipette** (with a pipette filler) to put 25.0 cm^3 of alkali into a **conical flask**.
3. Add a few drops of a **suitable indicator**. Stand the flask on a white tile so the colour change is easy to see.
4. Add acid from the burette while swirling the flask. Near the end, add it **drop by drop**.
5. Stop at the **end-point**: the indicator just changes colour permanently. Record the **final reading**. Titre = final reading − initial reading.
6. Do a rough titration first, then repeat accurately until you get **concordant results** (titres that agree closely, e.g. within 0.10 cm^3). Use the mean of the concordant titres.

Indicator	Colour in acid	Colour in alkali
methyl orange	red	yellow
thymolphthalein	colourless	blue
litmus	red	blue

Watch out

Don't use **universal indicator** in a titration: it changes colour gradually through many colours, so there is no sharp end-point. · The burette scale reads **downwards** (0.0 at the top). · A **volumetric** pipette, not a measuring cylinder, for the fixed volume.

2 Separation & Purification

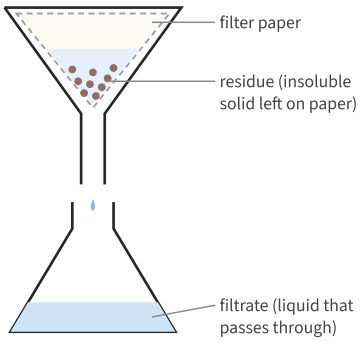
Definitions

Solvent – a substance that dissolves a solute.
Solute – a substance that is dissolved in a solvent.
Solution – a mixture of one or more solutes dissolved in a solvent.
Saturated solution – a solution containing the maximum concentration of a solute dissolved in the solvent at a specified temperature.
Residue – a substance that remains after evaporation, distillation, filtration or any similar process.
Filtrate – a liquid or solution that has passed through a filter.

Method	Separates	How it works	Example
Filtration	an **insoluble solid** from a liquid	The solid stays on the filter paper as the **residue**; the liquid passes through as the **filtrate**.	sand from water
Dissolving + filtration	a **soluble** solid from an **insoluble** one	Add a solvent that dissolves only one of them, stir, filter, then crystallise the filtrate.	salt from sand
Crystallisation	a **dissolved solid** from its solution	Heat to evaporate some solvent until the solution is saturated (crystals form on a glass rod dipped in and taken out). Leave to cool → crystals form. Filter, wash with a little cold solvent, dry between filter papers.	copper(II) sulfate crystals
Simple distillation	a **solvent** from a solution	The solvent boils, the vapour is cooled in the condenser and collected as the **distillate**. The solute stays in the flask.	pure water from sea water
Fractional distillation	**liquids that mix** but have **different boiling points**	The liquid with the **lowest boiling point** reaches the top of the fractionating column first and distils over while the thermometer reads its boiling point. The column gives a large surface area for repeated condensing and boiling.	ethanol from water; crude oil
Chromatography	a mixture of **soluble** substances (e.g. dyes)	Substances move up the paper different distances depending on their solubility in the solvent and attraction to the paper.	colours in ink

Choosing a method

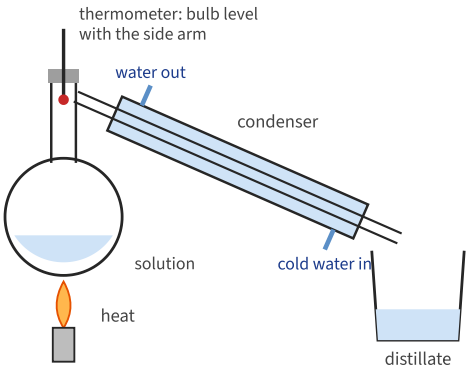
If you have…	Use
an insoluble solid in a liquid	filtration
a solid dissolved in a liquid, and you want the solid	crystallisation
a solid dissolved in a liquid, and you want the liquid	simple distillation
two (or more) liquids that mix, with different boiling points	fractional distillation
two solids, one soluble in a solvent and one not	dissolve, filter, crystallise
several soluble coloured (or colourless) substances	chromatography



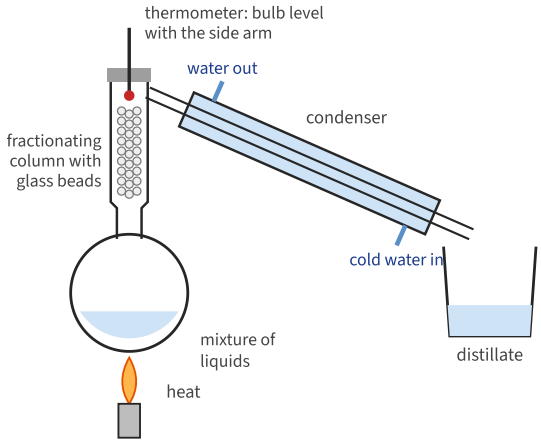
Filtration

Why crystallise, not evaporate to dryness?

Heating to dryness can decompose the product or drive off its water of crystallisation, and any soluble impurities would be left in the solid. Crystallising leaves most impurities behind in the solution.



Simple distillation

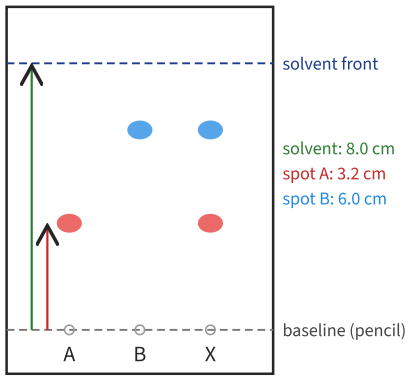


Fractional distillation

> ***Watch out — labelled-diagram marks***
> Thermometer bulb **level with the side arm** (it measures the vapour, not the liquid) · condenser water goes **in at the bottom, out at the top**, so the jacket stays full of cold water · the condenser must not be sealed at the end.

Chromatography

1. Draw a **baseline in pencil** (pencil doesn't dissolve in the solvent and run).
2. Put small spots of each sample on the line.
3. Stand the paper in a beaker with solvent **below the baseline** (so the spots don't dissolve into the solvent in the beaker). Put a lid on.
4. Remove the paper when the solvent is near the top; mark the **solvent front** and let it dry.
5. **Colourless** substances: spray with a **locating agent**, which makes the spots visible.



X contains A and B

Reading a chromatogram:

- A **pure** substance gives **one spot**; an **impure** one gives more than one.
- An unknown matches a known substance if its spot travels the **same distance** (same R_f) in the same solvent.
- A spot left on the baseline did not dissolve in that solvent.

> ***R_f value***
> R_f = distance travelled by substance ÷ distance travelled by solvent (both measured from the baseline)
> Spot A: $R_f = 3.2 \div 8.0 =$ **0.40** · Spot B: $R_f = 6.0 \div 8.0 =$ **0.75**
> R_f has no units and is always less than 1. Measure to the centre of the spot.

Purity

- A pure substance has a **sharp, fixed** melting point and boiling point (pure water: melts at 0 °C, boils at 100 °C).
- Impurities **lower** the melting point (and make it melt over a **range** of temperatures) and **raise** the boiling point.
- So: compare the measured melting/boiling point with the data-book value to identify a substance and to judge how pure it is.
- Purity matters for foods and medicines, where impurities could be harmful.

3 Identification of Ions & Gases

The practical papers print these tests at the back as "Notes for use in qualitative analysis", but **the theory papers don't** — so learn them. They are also the fastest marks in the exam once you know them.

Tests for aqueous cations

Add a few drops of the reagent, then **excess**. "Soluble in excess" = the precipitate dissolves when more reagent is added.

Cation	Aqueous sodium hydroxide	Aqueous ammonia
aluminium, Al^{3+}	white ppt., **soluble in excess**, giving a colourless solution	white ppt., insoluble in excess
ammonium, NH_4^+	ammonia produced on warming	—
calcium, Ca^{2+}	white ppt., insoluble in excess	no ppt. or very slight white ppt.
chromium(III), Cr^{3+}	green ppt., soluble in excess	green ppt., insoluble in excess
copper(II), Cu^{2+}	light blue ppt., insoluble in excess	light blue ppt., **soluble in excess, giving a dark blue solution**
iron(II), Fe^{2+}	green ppt., insoluble in excess, ppt. turns brown near the surface on standing	green ppt., insoluble in excess, ppt. turns brown near the surface on standing
iron(III), Fe^{3+}	red-brown ppt., insoluble in excess	red-brown ppt., insoluble in excess
zinc, Zn^{2+}	white ppt., **soluble in excess**, giving a colourless solution	white ppt., **soluble in excess**, giving a colourless solution

Telling the white precipitates apart (common question)

	Al^{3+}	Ca^{2+}	Zn^{2+}
excess NaOH(aq)	dissolves	stays	dissolves
excess NH_3(aq)	stays	(little or no ppt.)	dissolves

So: **Zn^{2+}** dissolves in excess of *both*; **Al^{3+}** only in NaOH; **Ca^{2+}** in neither (and gives little or no precipitate with ammonia). The flame test also confirms calcium (orange-red).

Flame tests

Cation	Flame colour
lithium, Li^+	red
sodium, Na^+	yellow
potassium, K^+	lilac
calcium, Ca^{2+}	orange-red
barium, Ba^{2+}	light green
copper(II), Cu^{2+}	blue-green

Doing a flame test

Clean a wire loop (nichrome or platinum) by dipping it in acid and holding it in a hot Bunsen flame until it gives no colour. Dip it in the sample, then hold it in the hot (blue) part of the flame and observe the colour.

Watch out

Lithium = **red**, calcium = **orange-red**: don't swap them. A dirty loop gives a false colour (sodium's yellow can hide other colours).

Tests for anions

Anion	Test	Result
carbonate, CO_3^{2-}	add dilute acid, then test for carbon dioxide gas	effervescence; carbon dioxide produced (turns limewater milky)
chloride, Cl^-	acidify with dilute **nitric acid**, then add aqueous **silver nitrate**	**white** ppt.
bromide, Br^-		**cream** ppt.
iodide, I^-		**yellow** ppt.
nitrate, NO_3^-	add aqueous sodium hydroxide, then aluminium foil; warm carefully	ammonia produced (turns damp red litmus paper blue)
sulfate, SO_4^{2-}	acidify with dilute **nitric acid**, then add aqueous **barium nitrate**	**white** ppt.
sulfite, SO_3^{2-}	add a small volume of acidified aqueous potassium manganate(VII)	potassium manganate(VII) changes from **purple to colourless**

Reasons examiners ask for

Why acidify first? To remove carbonate ions, which would also give a precipitate with silver or barium ions.

Why nitric acid? Hydrochloric acid would add Cl^- ions (false positive in the halide test); sulfuric acid would add SO_4^{2-} ions (false positive in the sulfate test).

Nitrate test: first check there's no NH_4^+, which gives ammonia with NaOH on its own.

Tests for gases

Gas	Test	Result
ammonia, NH_3	damp red litmus paper	turns **blue**
carbon dioxide, CO_2	bubble through limewater	limewater turns **milky**
chlorine, Cl_2	damp litmus paper	**bleached**
hydrogen, H_2	lighted splint	**'pops'**
oxygen, O_2	glowing splint	**relights**
sulfur dioxide, SO_2	acidified aqueous potassium manganate(VII)	turns from **purple to colourless**

> ***Watch out***
> Litmus must be **damp** (gases need to dissolve to act on it) · hydrogen: **lighted** splint; oxygen: **glowing** splint · say "limewater turns milky/cloudy", not "CO_2 turns milky" · tests for **water** (cobalt(II) chloride and copper(II) sulfate) are in Topic 10.

Self-check before the topic questions

- ☐ Name the apparatus for measuring time, temperature, mass and volume, and compare pipette, burette, measuring cylinder and gas syringe
- ☐ Describe an acid–base titration step by step, including the end-point and concordant results
- ☐ Define solvent, solute, solution, saturated solution, residue and filtrate
- ☐ Describe filtration, crystallisation, simple and fractional distillation, and choose the right method for a mixture
- ☐ Label a distillation apparatus (thermometer position, condenser water flow)
- ☐ Run and interpret a chromatogram, including locating agents, and calculate R_f
- ☐ Use melting and boiling points to identify a substance and judge its purity
- ☐ Give the results for all 8 cations with NaOH(aq) and NH_3(aq), and all 6 flame colours
- ☐ Describe the tests for carbonate, chloride, bromide, iodide, nitrate, sulfate and sulfite
- ☐ Describe the tests for ammonia, carbon dioxide, chlorine, hydrogen, oxygen and sulfur dioxide

Practical Skills (Paper 6)

Cambridge IGCSE Chemistry 0620 · Alternative to Practical · Use with Topic 12 and real Paper 6 past papers

1 What Paper 6 asks

Paper 6 is a written paper (1 hour, 40 marks) about experiments you could do in a lab. The same few question types come up every year:

- **Reading apparatus** from diagrams to complete a table
- **Results and graphs:** complete tables, plot, best-fit line, read values
- **Conclusions:** describe the pattern, explain it with theory
- **Observations / tests:** what you'd see; identify ions and gases
- **Evaluation:** problems with a method and improvements
- **Planning:** write a method (usually the last question)

The qualitative analysis tests (Topic 12) are printed at the back of the paper — but you still need to know *how* to describe the observations precisely.

2 Variables and planning

Independent variable	Dependent variable	Control variables
the one you deliberately **change**	the one you **measure**	everything else you **keep the same**, so the test is fair

A plan that scores: include all of these

1. Numbered steps a classmate could follow, naming **apparatus and quantities** ("use a measuring cylinder to add 50 cm^3 of dilute hydrochloric acid to a conical flask").
2. What you **change**, and a sensible range (at least 5 different values).
3. What you **measure**, and with which apparatus.
4. What you **keep the same** (name at least two control variables).
5. **Repeat** and take the mean.
6. How the results answer the question ("the shorter the time, the faster the rate").

Worked example

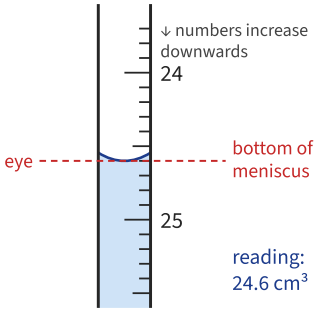
Plan an investigation into how the concentration of hydrochloric acid affects the rate of its reaction with magnesium ribbon.

1. Measure 50 cm^3 of 0.5 mol/dm^3 hydrochloric acid with a measuring cylinder and pour it into a conical flask.
2. Connect a gas syringe to the flask with a bung and delivery tube.
3. Add a 3 cm length of magnesium ribbon, quickly replace the bung and start the stopwatch.
4. Record the volume of hydrogen in the syringe every 10 seconds until the reaction stops.
5. Repeat with acid of 1.0, 1.5, 2.0 and 2.5 mol/dm^3.
6. Keep the same: volume of acid, length of magnesium, temperature.
7. Repeat each concentration and calculate the mean. The greater the volume of gas in the first 30 s (or the steeper the graph), the faster the rate.

3 Apparatus and reading scales

Apparatus	Reading it
burette	scale reads **downwards**; divisions of 0.1 cm^3; record to 1 decimal place (e.g. 0.0, 24.6)
volumetric pipette	one fixed volume, e.g. 25.0 cm^3
measuring cylinder	read the bottom of the meniscus at eye level
thermometer	read at eye level; usually to the nearest 0.5 °C or 1 °C, depending on the scale
gas syringe	read where the plunger is
stopwatch	convert minutes to seconds: 1 min 25 s = **85 s**
balance	record all the digits shown, e.g. 2.45 g

Watch out
Burette: reading **upwards** by mistake (24.6 read as 23.4) is the classic error · titre = final − initial · record every reading in a column to the **same** number of decimal places.



Reading a burette

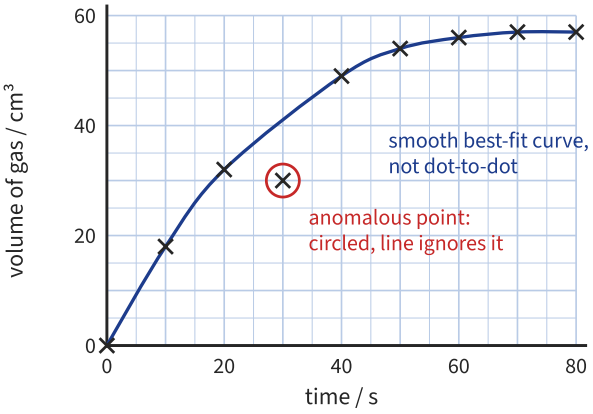
4 Results tables

- Each column heading has the **quantity and unit**, e.g. *temperature / °C* or *time (s)*.
- **No units** in the body of the table — just numbers.
- Same number of decimal places down each column, matching the apparatus.
- Include calculated columns (change in temperature, titre, mean).

initial temperature / °C	final temperature / °C	temperature change / °C
21.0	28.5	7.5
21.5	32.0	10.5

5 Graphs

- **Independent** variable on the **x-axis**; dependent on the y-axis.
- Label both axes with **quantity and unit** ("time / s").
- Choose a **linear** scale that is easy to read (each square = 1, 2, 5 or 10 units — never 3) and makes the points fill **at least half** of the grid in each direction.
- Plot points **accurately** with small neat crosses.
- Draw a **smooth curve or a ruled straight line** of best fit — never join the dots.
- **Anomalous** results: circle them and leave them out of the line.
- When reading a value off the graph, **draw the lines** on the graph to show how you did it.



Volume of hydrogen against time

Describing and explaining a trend

Describe: use the data — "as the concentration increases, the time taken decreases"; quote numbers from the table or graph.

Explain: use the theory — e.g. "more particles per unit volume, so more frequent collisions".

Reaction finished? On a gas-volume graph the curve levels off: one reactant has been used up.

6 Describing observations precisely

Write what you would **see**, not what you know is happening.

Situation	Say	Don't say
gas produced	bubbles / fizzing / effervescence	"a gas was given off", "hydrogen was produced" (a conclusion, not an observation)
colour change	"the solution changed from **blue to colourless**" (both colours)	"it changed colour", "it went colourless"
no colour	**colourless**	"clear" (clear means see-through; a blue solution is clear)
a solid forms in a solution	"a **white precipitate**" (give the colour)	"white solution", "it went cloudy" (except limewater)
solid used up	"the solid disappears" / "dissolves"	"the solid melts"
temperature change	"the flask gets warmer"; give readings	"it reacts"
flame test	the exact colour: lilac, orange-red…	"purple", "orange" for calcium

7 Errors and improvements

Problem	Improvement
Heat lost to the surroundings in a temperature-change experiment	Use an insulated (polystyrene) cup with a lid
Gas escapes before the bung is put in	Put the solid in a small tube inside the flask, insert the bung, then tip it over to start the reaction
Acid spray escapes in a mass-loss experiment	Put a cotton wool plug in the neck of the flask (gas gets out, spray doesn't)
Volume measured with a measuring cylinder	Use a burette or volumetric pipette (more precise)
Gas is soluble in water so can't be collected over water	Collect it in a gas syringe
Hard to judge the end-point / when a cross disappears	Use a light sensor (or the same person judges every time)
Temperature changes between runs	Use a water bath to keep the temperature constant
Reaction too fast to time accurately	Use more dilute acid or larger pieces of solid
Only one set of results	Repeat, identify anomalies and take the mean

Watch out — answers that score nothing

"Human error" · "use more accurate equipment" without naming it · "do it more carefully" · "to make it a fair test" without saying what you keep the same. Always name the **specific** problem and the **specific** fix.

8 Safety

Eye protection is always expected. Beyond that, the precaution must match the **specific hazard**:

Hazard	Precaution
Flammable liquids (ethanol, hexane)	No naked flames; heat with a water bath or electric heater
Toxic gases (chlorine, sulfur dioxide)	Use a fume cupboard
Corrosive acids and alkalis (e.g. sodium hydroxide)	Wear eye protection (and gloves); wash spills with plenty of water
Heating a solid in a test tube	Point the open end away from people
Collecting a gas over water while heating	Remove the delivery tube from the water before you stop heating, so water isn't sucked back into the hot tube
Hot apparatus	Let it cool; hold with tongs

Self-check before Paper 6 past papers

☐ Identify the independent, dependent and control variables in any experiment

☐ Write a full plan: steps with apparatus and quantities, range, measurements, controls, repeats, conclusion

☐ Read a burette, thermometer, measuring cylinder, gas syringe and stopwatch from a diagram

☐ Draw a results table with units in the headings and consistent decimal places

☐ Plot a graph with a good scale, a line of best fit and anomalous points identified

☐ Describe and explain a trend using numbers from the data

☐ Describe observations precisely (bubbles, both colours, colourless, precipitate colour)

☐ Give a specific problem with a method and a specific improvement

☐ Match a safety precaution to a specific hazard